TP Veeam : Mise en place de la sauvegarde et restauration avec Veeam Backup & Replication 12.3

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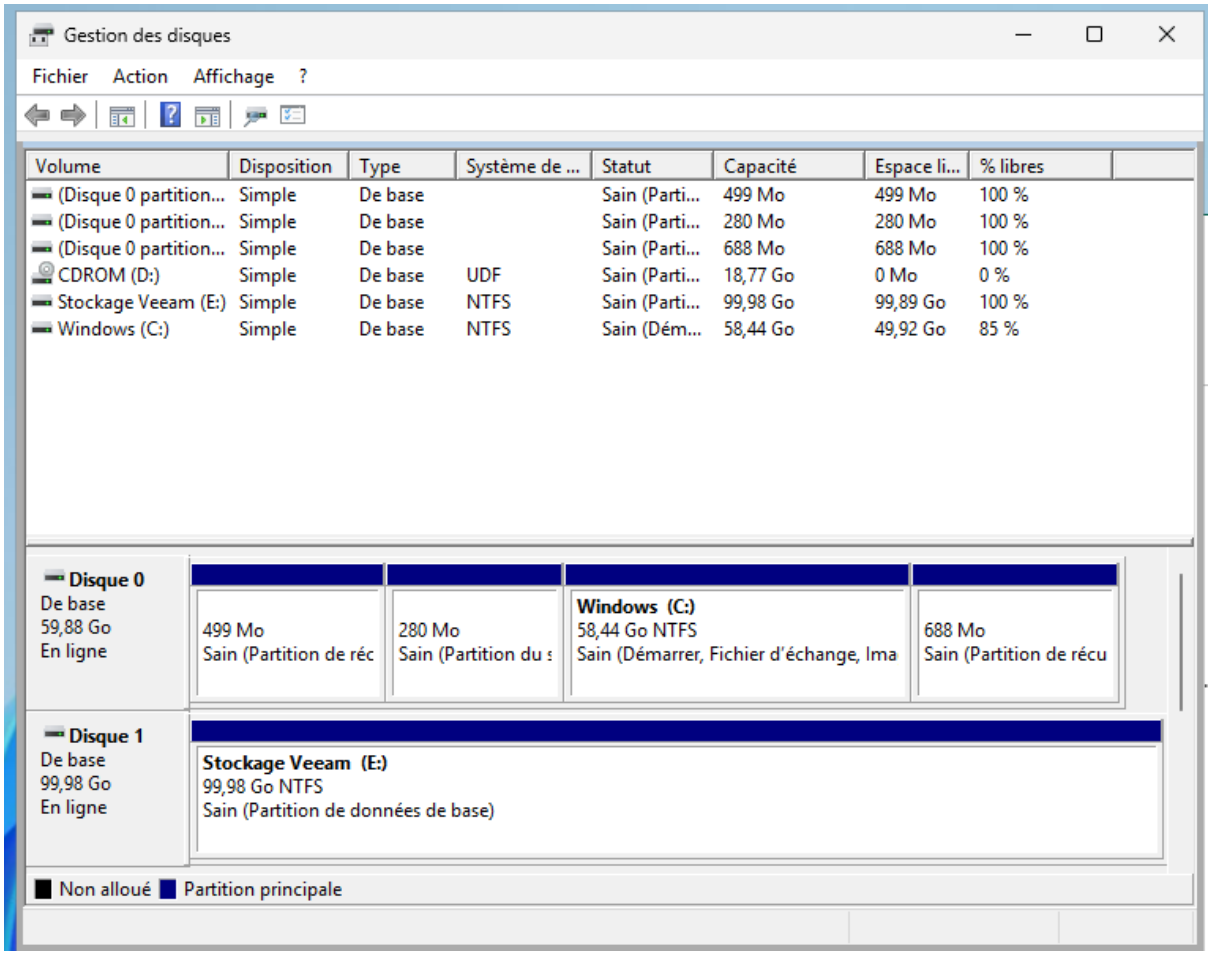
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Introduction

Ce TP a pour but de nous familiariser avec l'installation de Veeam et de Windows Server. Veeam est un logiciel de gestion de sauvegarde de parc informatique. Nous allons dans un premier temps installer Windows Server, car il s'agit de la base pour que nous puissions travailler. Nous installerons ensuite Veeam via un iso, directement sur Windows Server.

ÉTAPE 1 : Installer Veeam Backup & Replication sur le serveur



On créé le volume pour le disque dur de 100 Go soit exploitable.

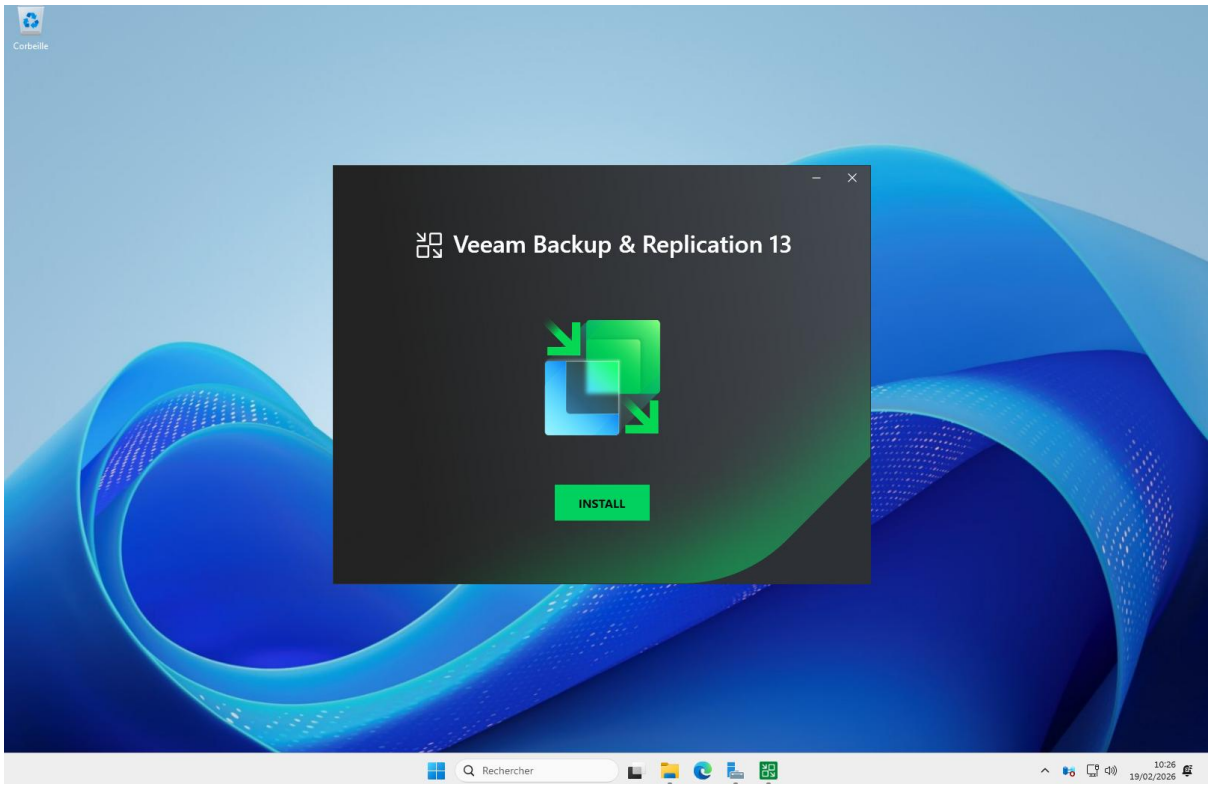
On lance l'installation de Veeam Backup & Replication 13.

On choisit les options par défaut et on accepte l'installation de *PostgreSQL* comme base de données.

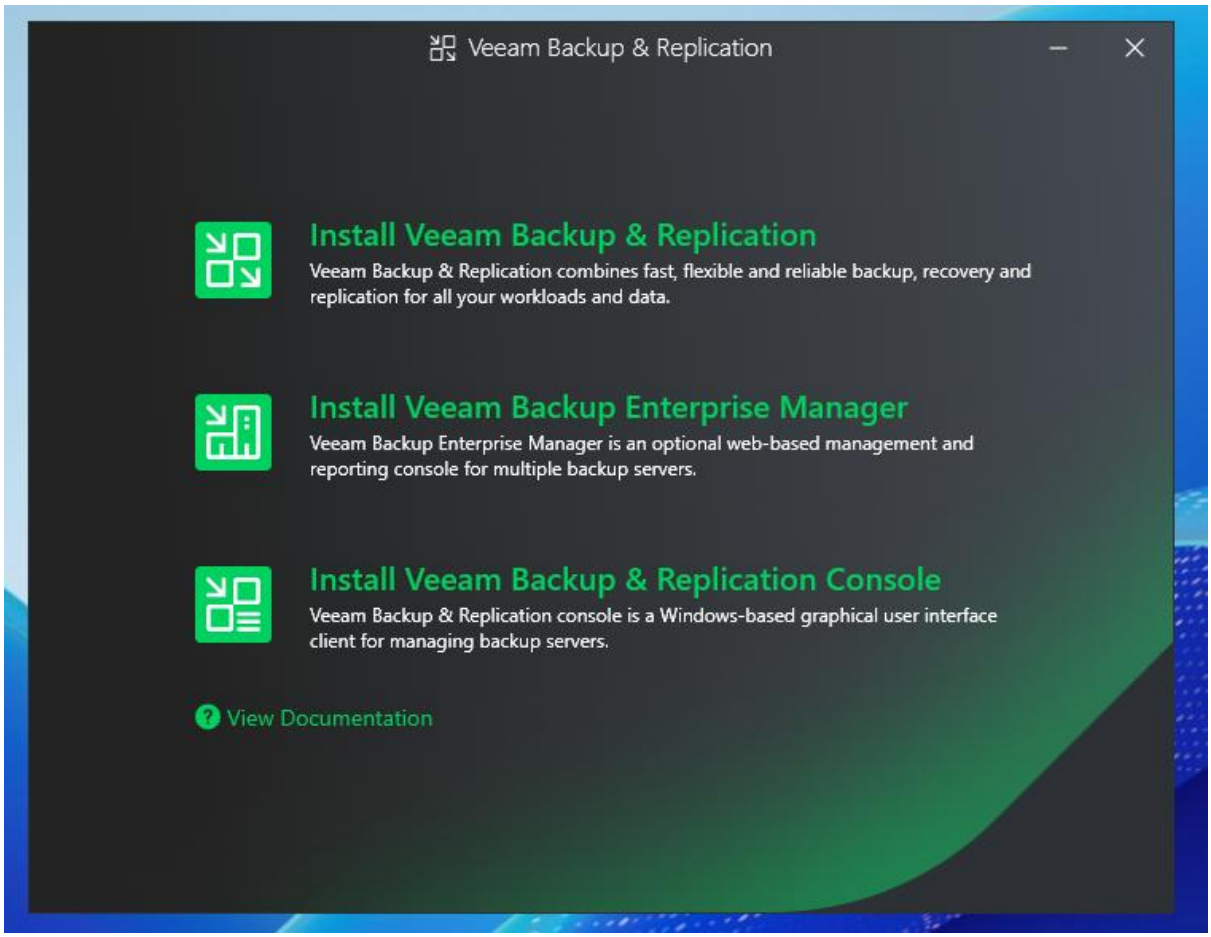
Il faut laisser les ports proposés :

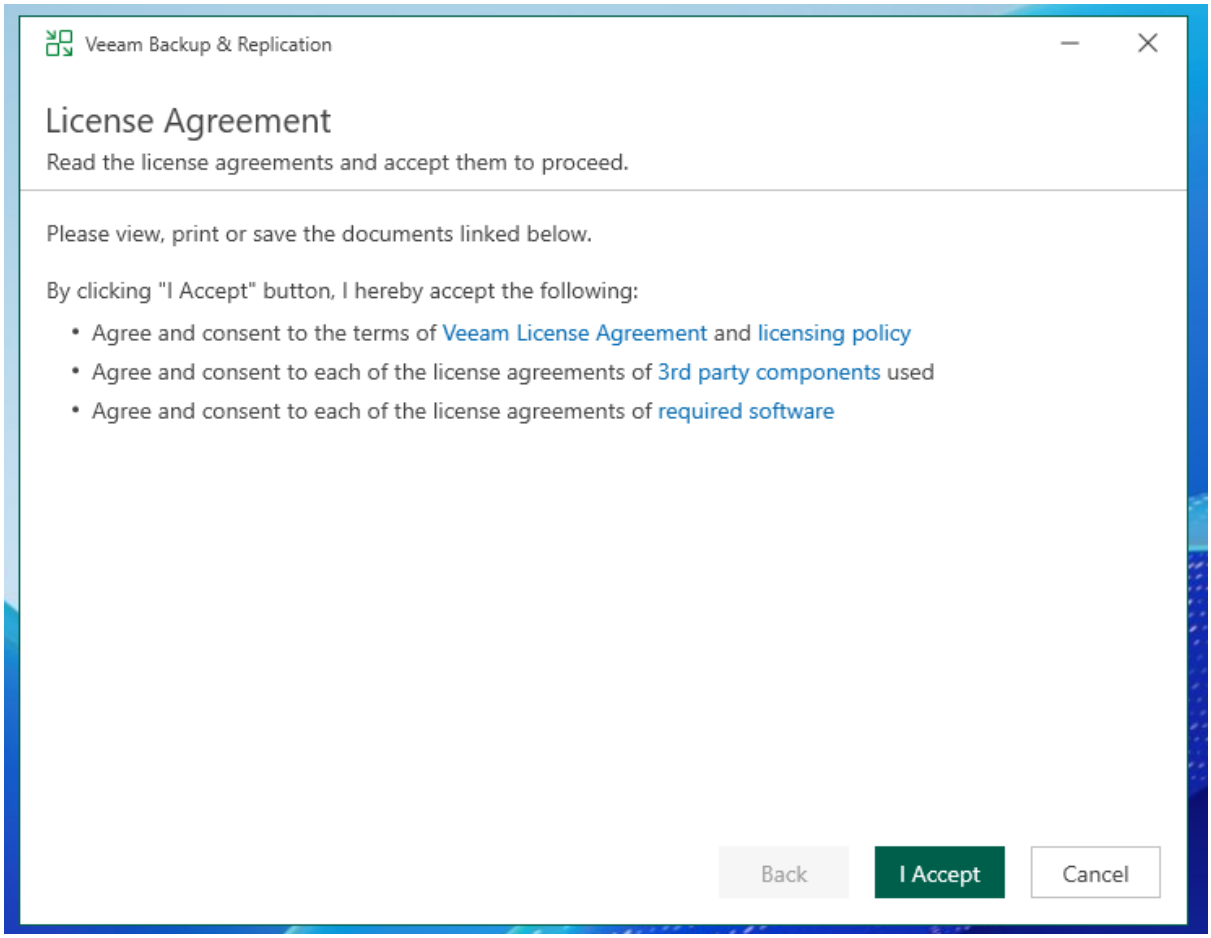
Port principal : 9392

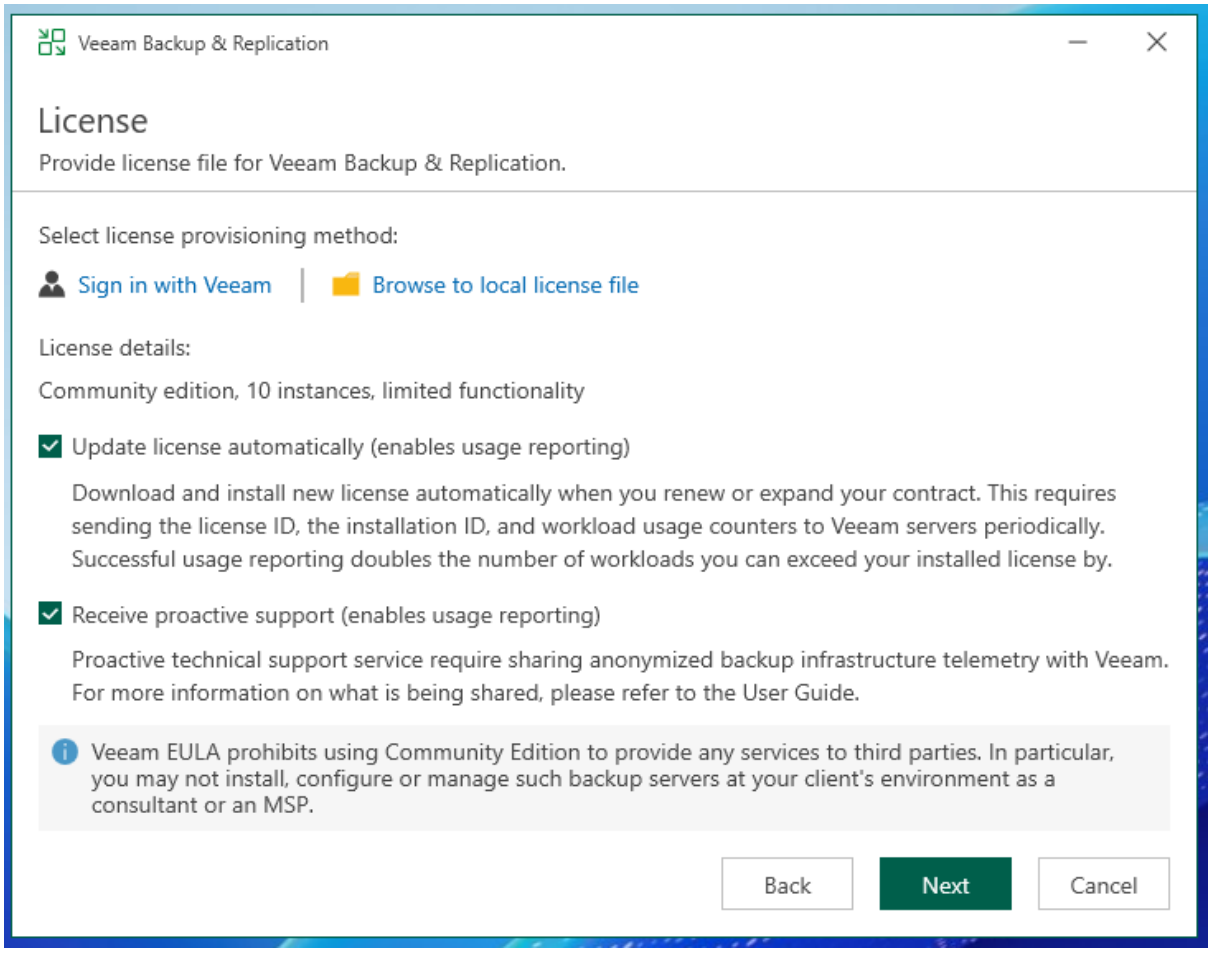
REST API : 9419

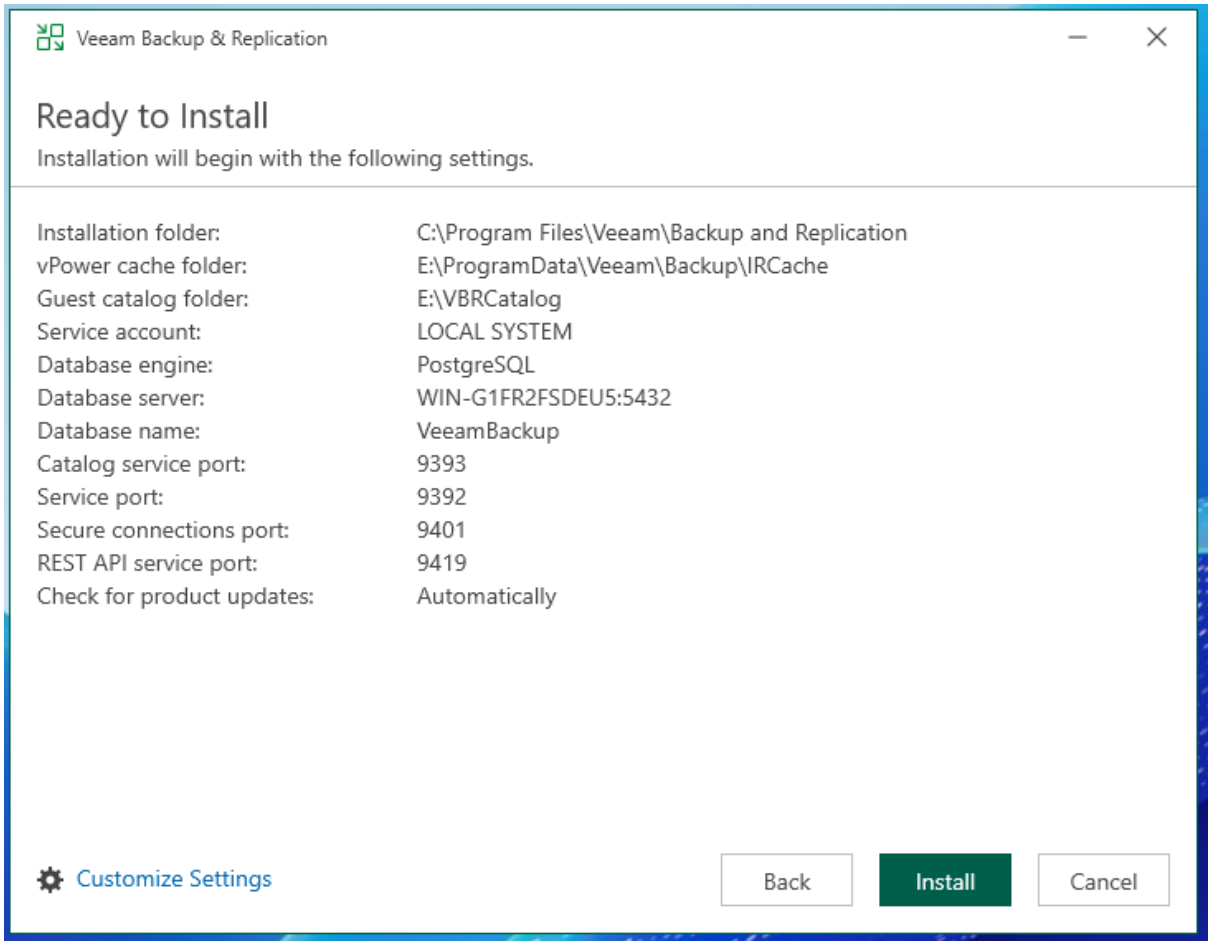


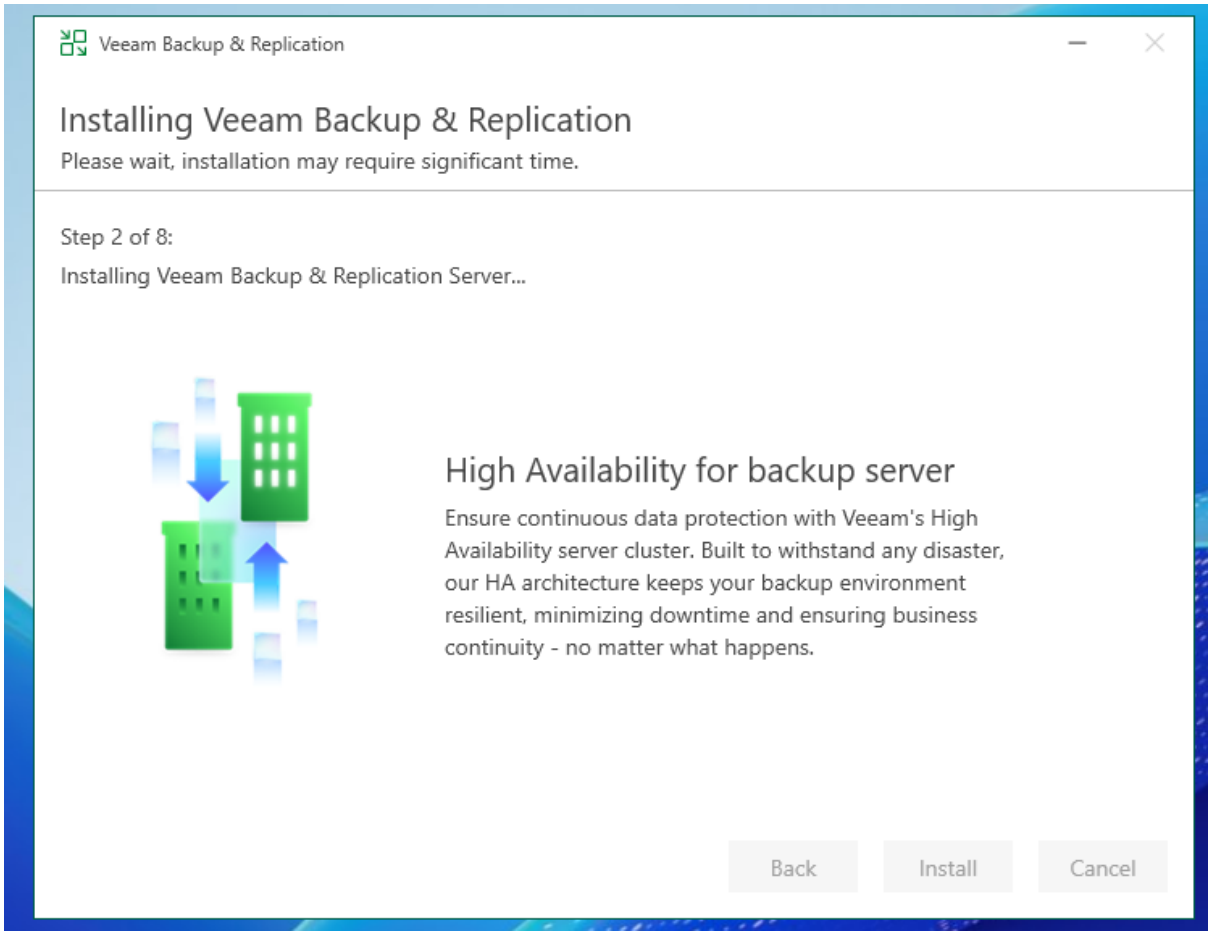
On sélectionne *Install Veeam Backup & Replication*.



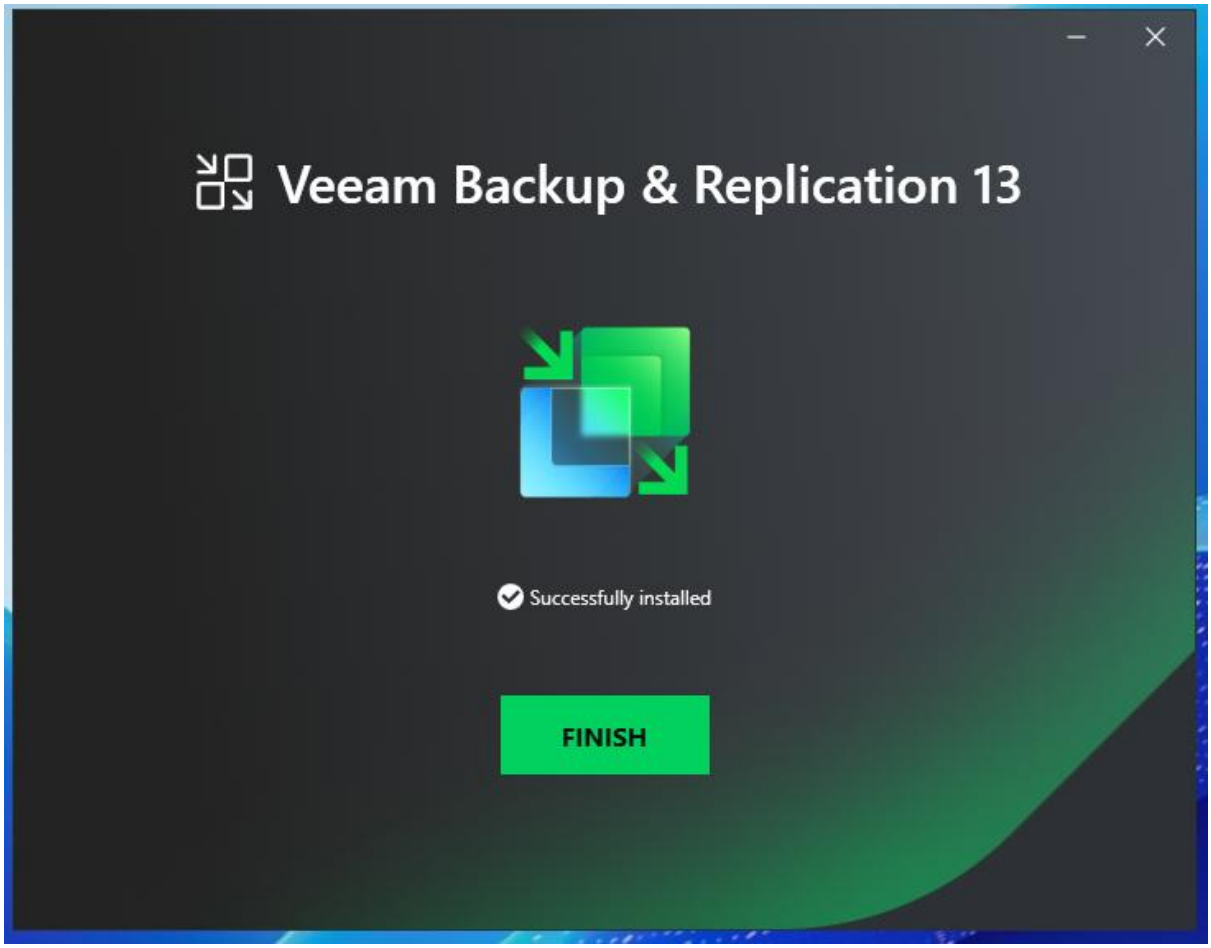

Veeam Backup & Replication
License Agreement
Read the license agreements and accept them to proceed.
Please view, print or save the documents linked below.
By clicking "I Accept" button, I hereby accept the following:
Agree and consent to the terms of Veeam License Agreement and licensing policy
Agree and consent to each of the license agreements of 3rd party components used
Agree and consent to each of the license agreements of required software
Back
I Accept
Cancel

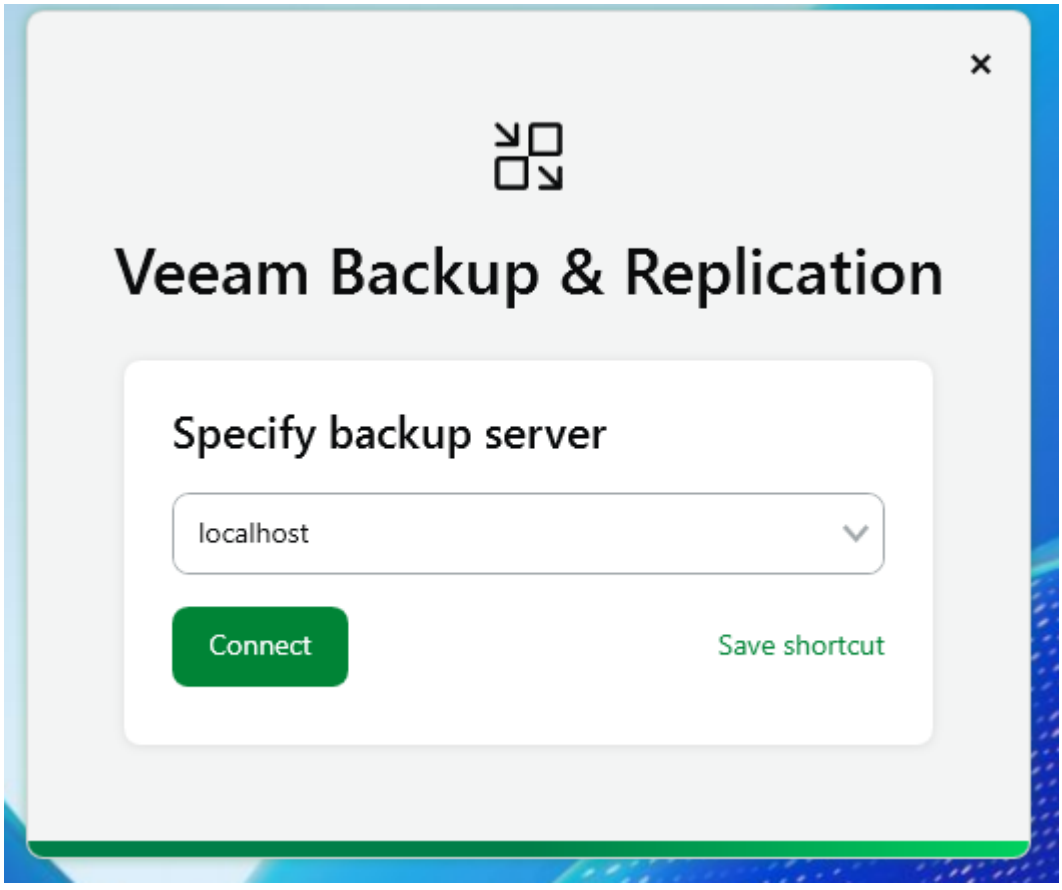

Veeam Backup & Replication
License
Provide license file for Veeam Backup & Replication.
Select license provisioning method:
Sign in with Veeam
Browse to local license file
License details:
Community edition, 10 instances, limited functionality
Update license automatically (enables usage reporting)
Download and install new license automatically when you renew or expand your contract. This requires sending the license ID, the installation ID, and workload usage counters to Veeam servers periodically. Successful usage reporting doubles the number of workloads you can exceed your installed license by.
Receive proactive support (enables usage reporting)
Proactive technical support service require sharing anonymized backup infrastructure telemetry with Veeam. For more information on what is being shared, please refer to the User Guide.
Veeam EULA prohibits using Community Edition to provide any services to third parties. In particular, you may not install, configure or manage such backup servers at your client's environment as a consultant or an MSP.
Back
Next
Cancel


Veeam Backup & Replication
Ready to Install
Installation will begin with the following settings.
Installation folder: C:\Program Files\Veeam\Backup and Replication
vPower cache folder: E:\ProgramData\Veeam\Backup\IRCache
Guest catalog folder: E:\VBRCatalog
Service account: LOCAL SYSTEM
Database engine: PostgreSQL
Database server: WIN-G1FR2FSDEU5:5432
Database name: VeeamBackup
Catalog service port: 9393
Service port: 9392
Secure connections port: 9401
REST API service port: 9419
Check for product updates: Automatically
Customize Settings
Back
Install
Cancel



L'installation peut prendre une trentaine de minutes.


Veeam Backup & Replication 13
Successfully installed
FINISH


Veeam Backup & Replication
Specify backup server
localhost
Connect
Save shortcut

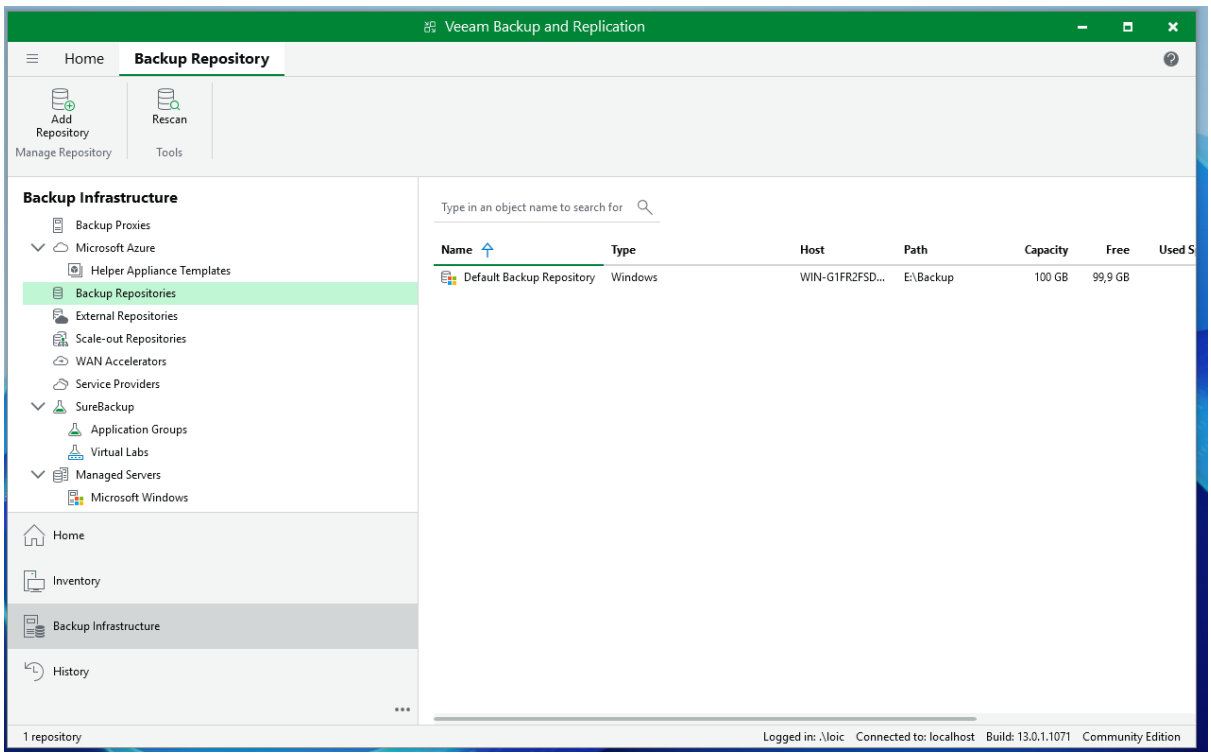
ÉTAPE 2 : Créer un dépôt de sauvegarde

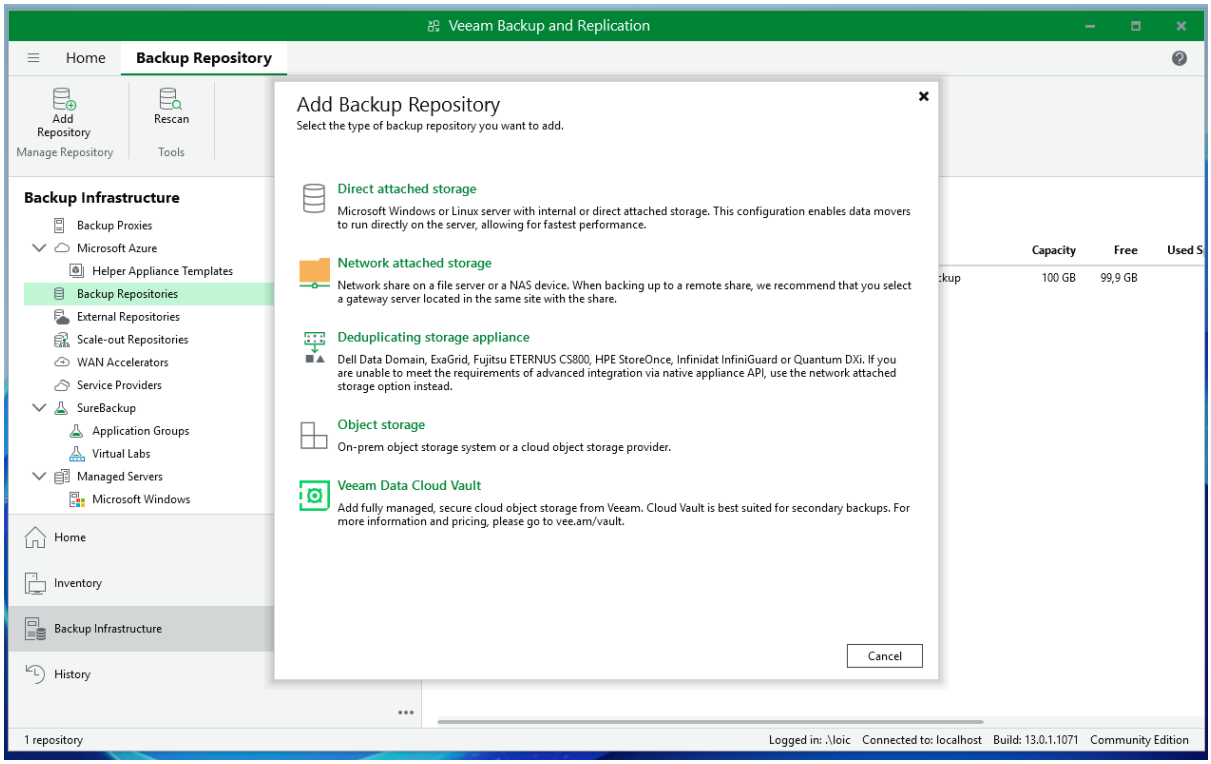
Dans la console Veeam, il faut aller dans *Backup Infrastructure* > *Backup Repositories*

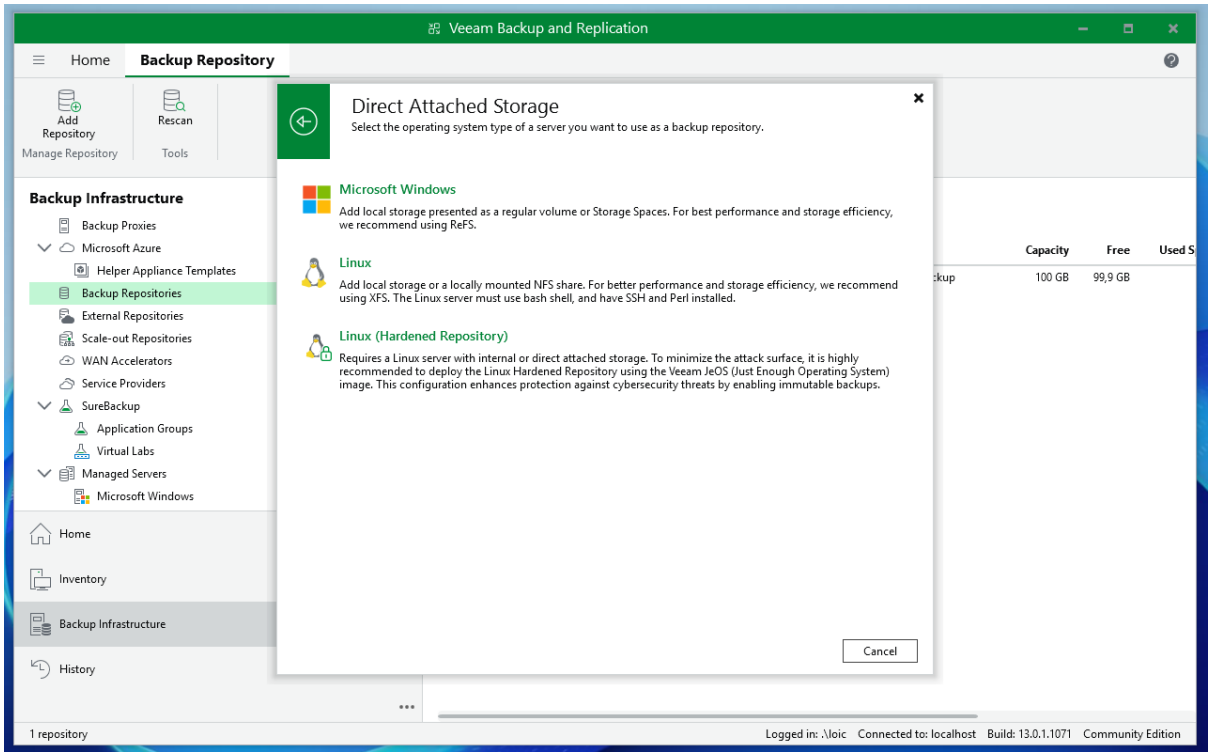
On ajoute un nouveau dépôt, nommé *Sauvegarde*.

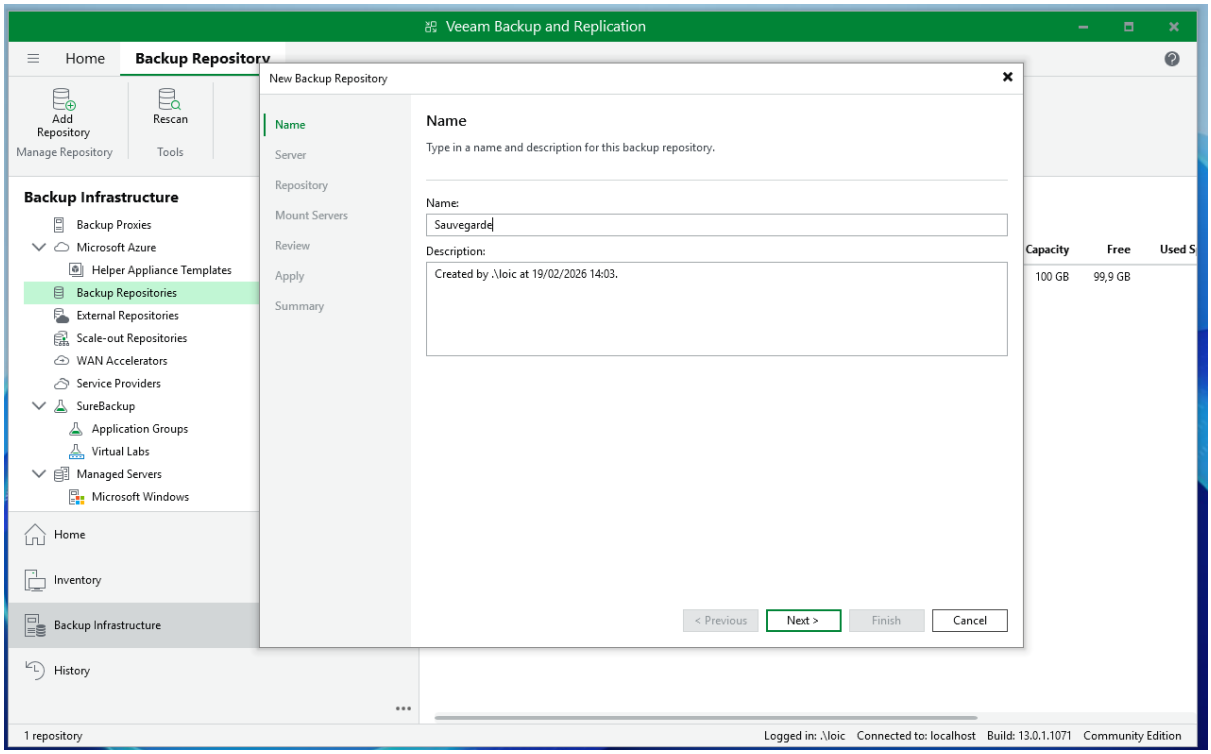
On choisit un dossier sur le disque E: (ex. E:\Sauvegarde).

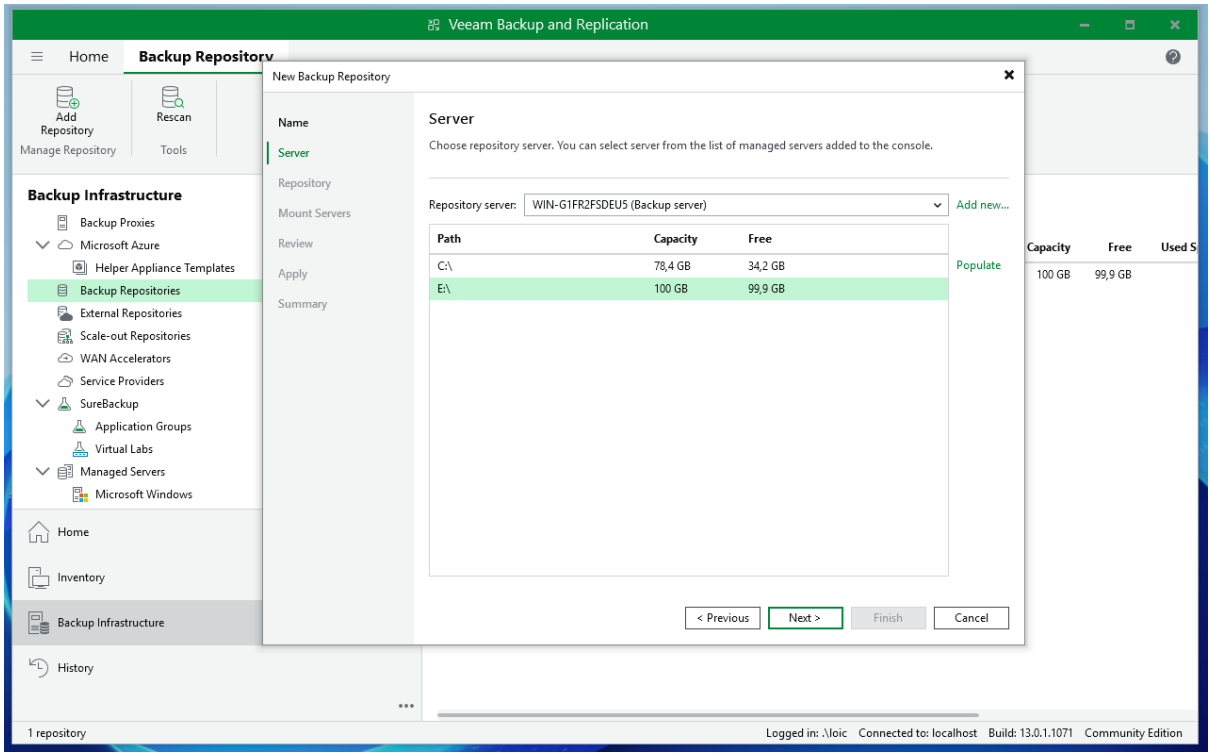
Il faut laisser les options par défaut et activer le support vPower NFS si proposé.

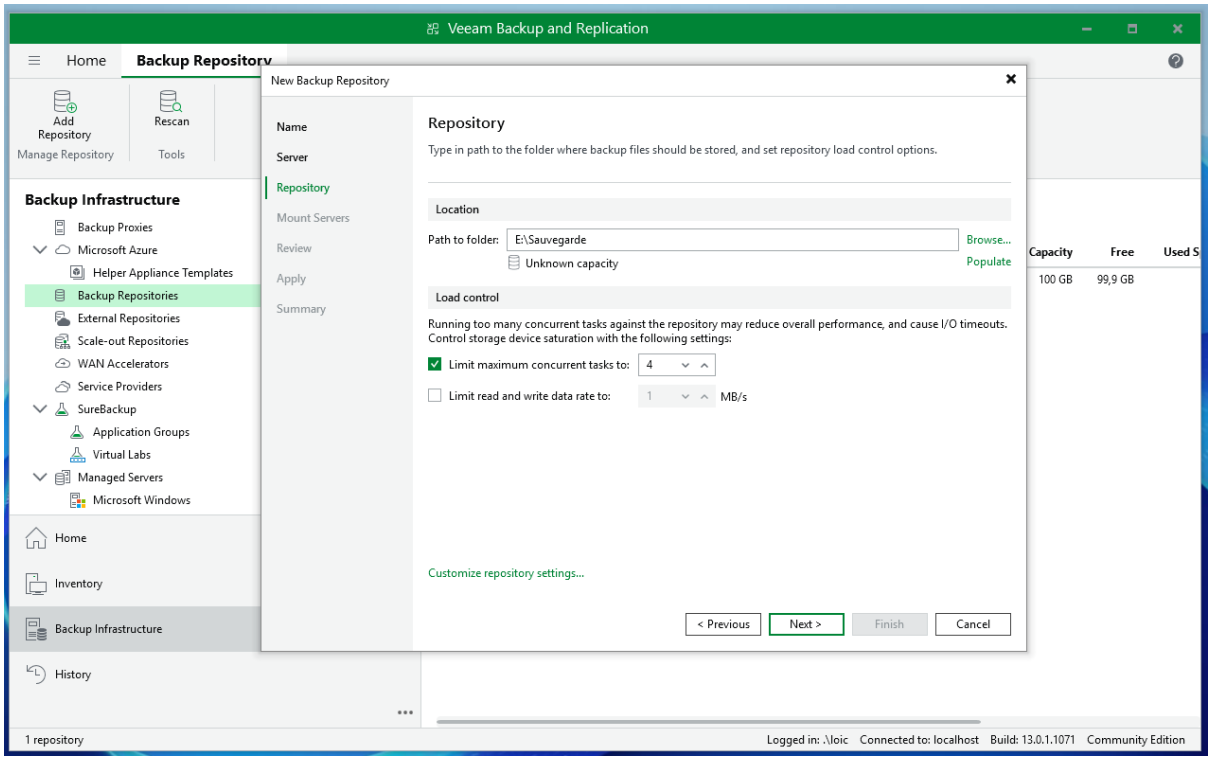


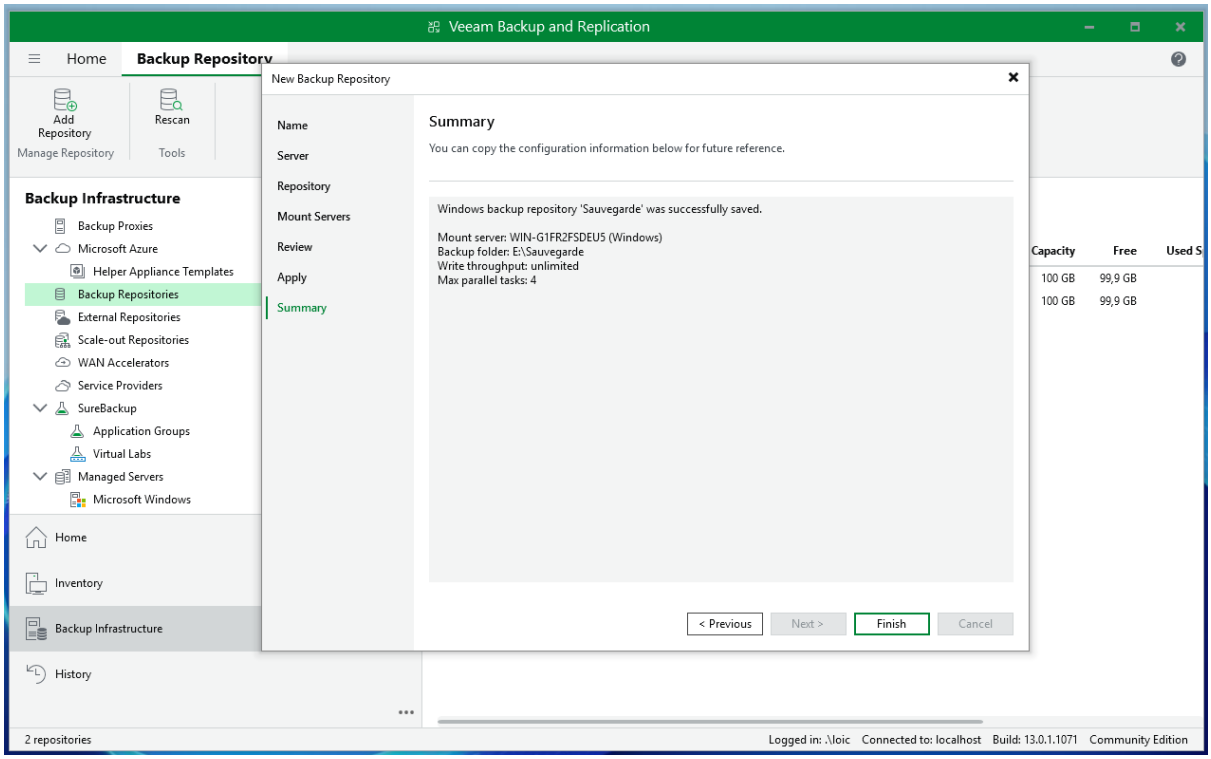

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Service Providers
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History
Capacity
Free
Used S
100 GB
99,9 GB
Add Backup Repository
Select the type of backup repository you want to add.
Direct attached storage
Microsoft Windows or Linux server with internal or direct attached storage. This configuration enables data movers to run directly on the server, allowing for fastest performance.
Network attached storage
Network share on a file server or a NAS device. When backing up to a remote share, we recommend that you select a gateway server located in the same site with the share.
Deduplicating storage appliance
Dell Data Domain, ExaGrid, Fujitsu ETERNUS CS800, HPE StoreOnce, Infinidat InfiniGuard or Quantum DXi. If you are unable to meet the requirements of advanced integration via native appliance API, use the network attached storage option instead.
Object storage
On-prem object storage system or a cloud object storage provider.
Veeam Data Cloud Vault
Add fully managed, secure cloud object storage from Veeam. Cloud Vault is best suited for secondary backups. For more information and pricing, please go to vee.am/vault.
Cancel
1 repository
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

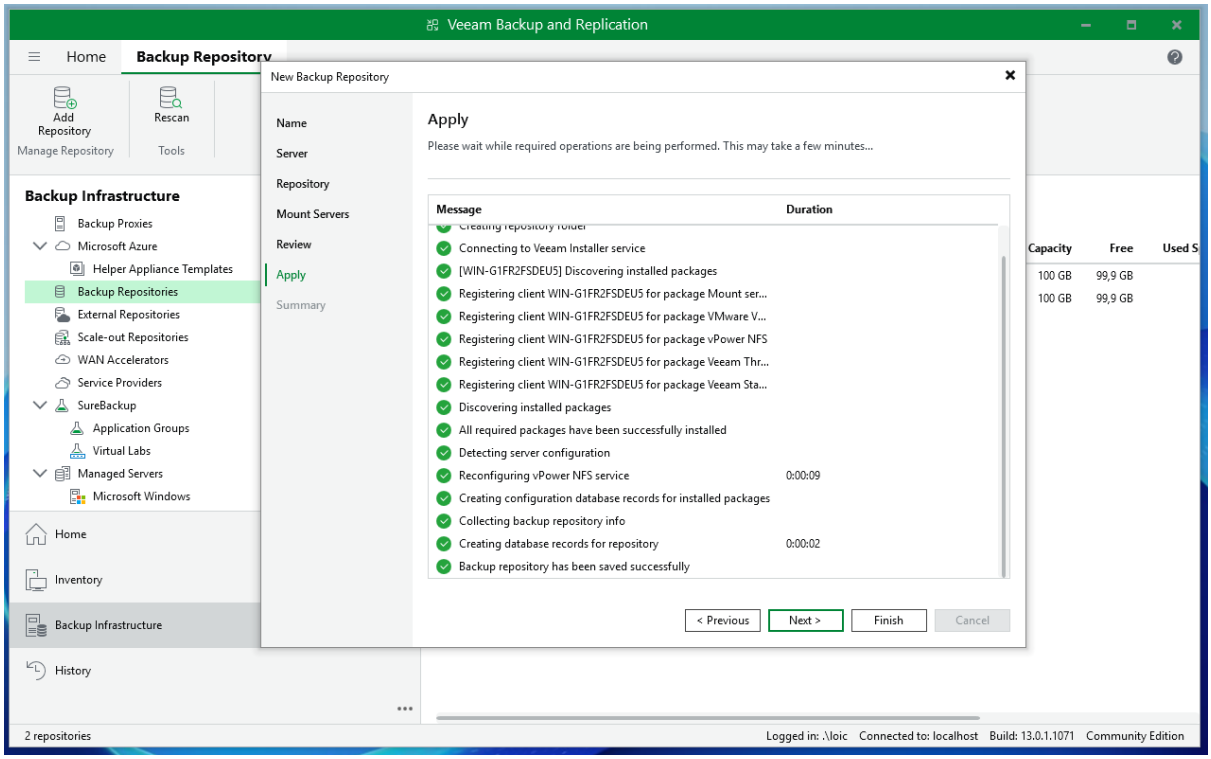

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Direct Attached Storage
Select the operating system type of a server you want to use as a backup repository.
Microsoft Windows
Add local storage presented as a regular volume or Storage Spaces. For best performance and storage efficiency, we recommend using ReFS.
Linux
Add local storage or a locally mounted NFS share. For better performance and storage efficiency, we recommend using XFS. The Linux server must use bash shell, and have SSH and Perl installed.
Linux (Hardened Repository)
Requires a Linux server with internal or direct attached storage. To minimize the attack surface, it is highly recommended to deploy the Linux Hardened Repository using the Veeam JeOS (Just Enough Operating System) image. This configuration enhances protection against cybersecurity threats by enabling immutable backups.
Cancel
1 repository
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

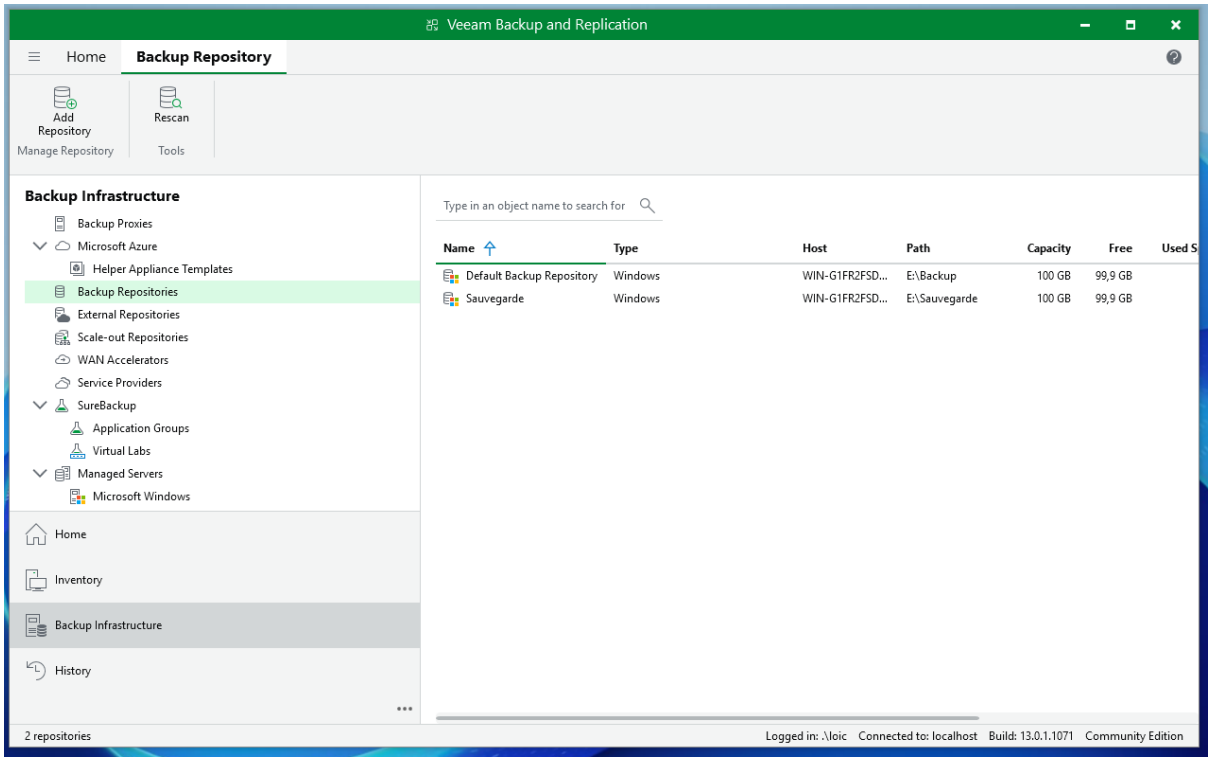

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1 repository
New Backup Repository
Name
Server
Repository
Mount Servers
Review
Apply
Summary
Name
Type in a name and description for this backup repository.
Name:
Sauvegarde
Description:
Created by .\loic at 19/02/2026 14:03.
< Previous
Next >
Finish
Cancel
Capacity
Free
Used S
100 GB
99,9 GB
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
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1 repository
New Backup Repository
Name
Server
Repository
Mount Servers
Review
Apply
Summary
Server
Choose repository server. You can select server from the list of managed servers added to the console.
Repository server:
WIN-G1FR2FSDEU5 (Backup server)
Add new...
Path
Capacity
Free
C:\
78,4 GB
34,2 GB
E:\
100 GB
99,9 GB
Populate
< Previous
Next >
Finish
Cancel
Capacity
Free
Used S
100 GB
99,9 GB
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
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1 repository
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition
New Backup Repository
Name
Server
Repository
Mount Servers
Review
Apply
Summary
Repository
Type in path to the folder where backup files should be stored, and set repository load control options.
Location
Path to folder:
E:\Sauvegarde
Browse...
Unknown capacity
Populate
Load control
Running too many concurrent tasks against the repository may reduce overall performance, and cause I/O timeouts.
Control storage device saturation with the following settings:
Limit maximum concurrent tasks to:
4
Limit read and write data rate to:
1
MB/s
Customize repository settings...
< Previous
Next >
Finish
Cancel
Capacity
Free
Used S
100 GB
99,9 GB

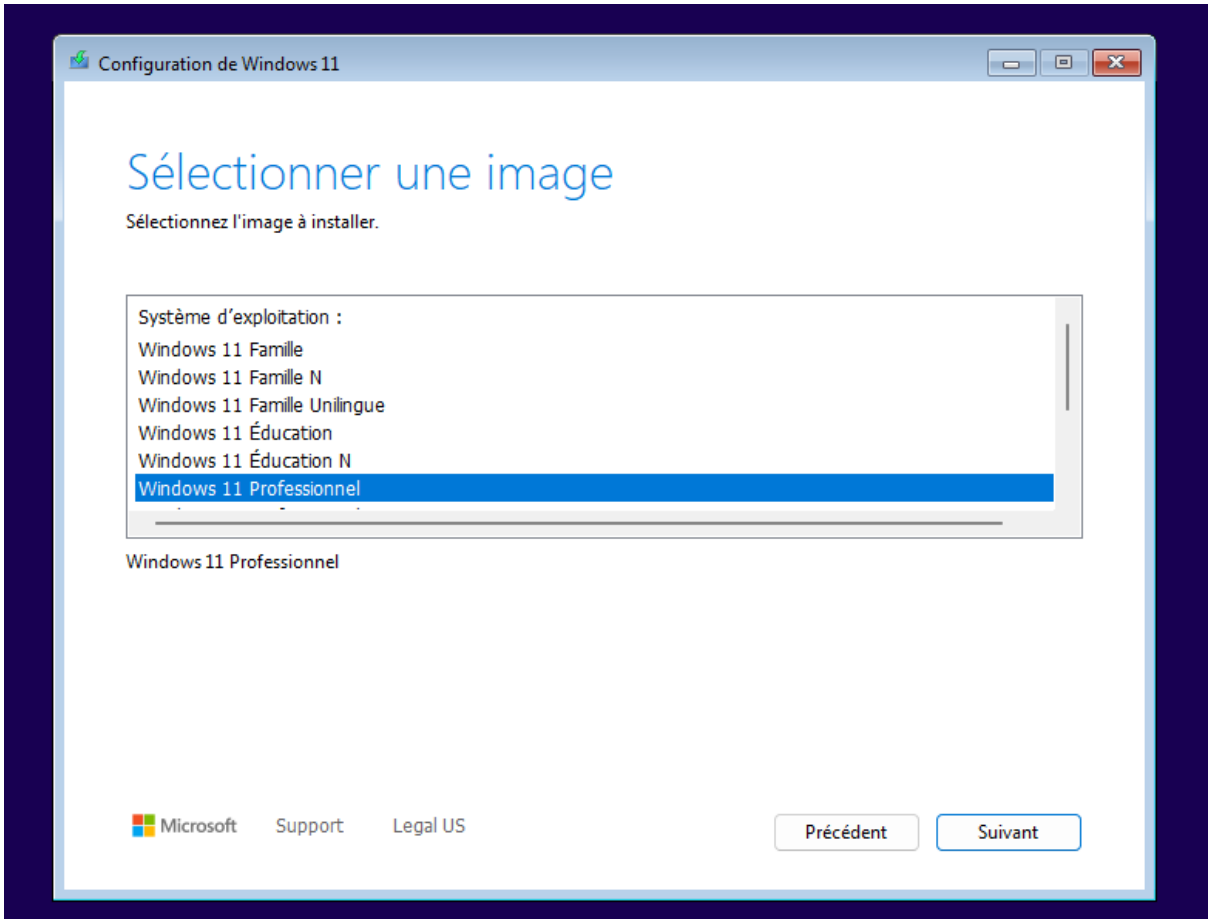

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2 repositories
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition
New Backup Repository
Name
Server
Repository
Mount Servers
Review
Apply
Summary
Summary
You can copy the configuration information below for future reference.
Windows backup repository 'Sauvegarde' was successfully saved.
Mount server: WIN-G1FR2FSDEU5 (Windows)
Backup folder: E:\Sauvegarde
Write throughput: unlimited
Max parallel tasks: 4
< Previous
Next >
Finish
Cancel
Capacity
Free
Used S
100 GB
99,9 GB
100 GB
99,9 GB

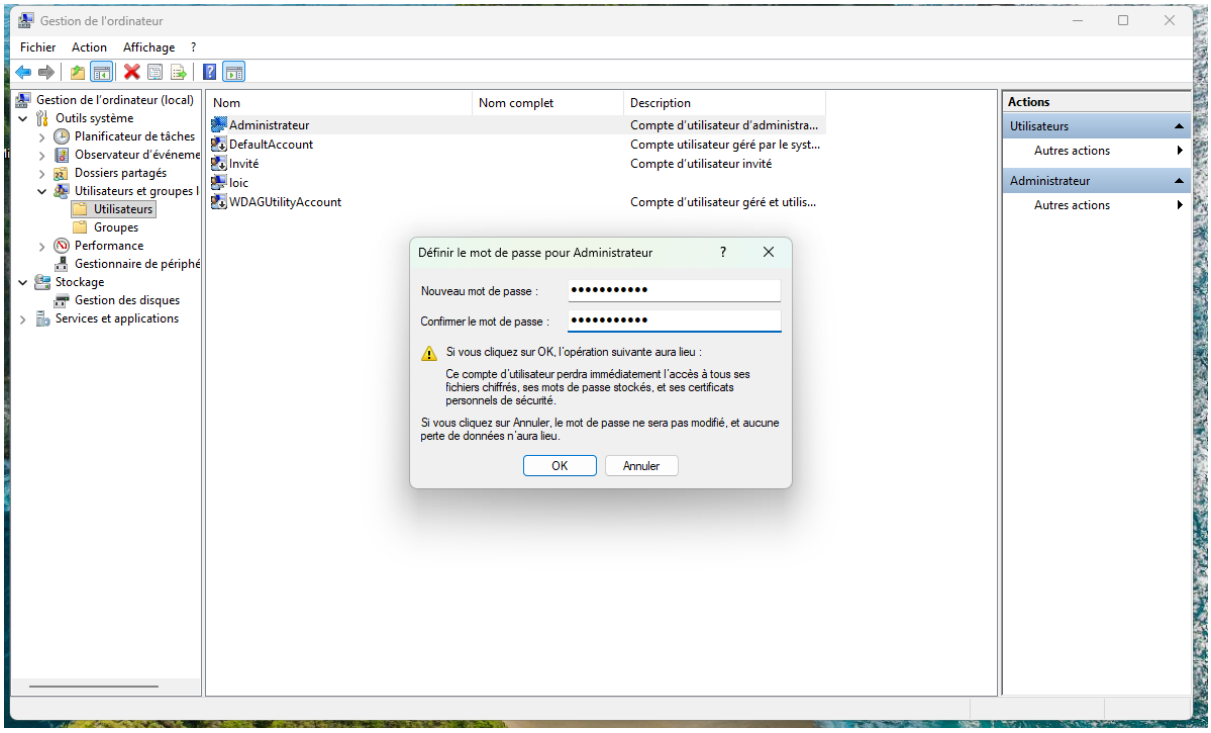

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Name
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Mount Servers
Review
Apply
Summary
Apply
Please wait while required operations are being performed. This may take a few minutes...
Message
Duration
Connecting to Veeam Installer service
[WIN-G1FR2FSDEU5] Discovering installed packages
Registering client WIN-G1FR2FSDEU5 for package Mount ser...
Registering client WIN-G1FR2FSDEU5 for package VMware V...
Registering client WIN-G1FR2FSDEU5 for package vPower NFS
Registering client WIN-G1FR2FSDEU5 for package Veeam Thr...
Registering client WIN-G1FR2FSDEU5 for package Veeam Sta...
Discovering installed packages
All required packages have been successfully installed
Detecting server configuration
Reconfiguring vPower NFS service 0:00:09
Creating configuration database records for installed packages
Collecting backup repository info
Creating database records for repository 0:00:02
Backup repository has been saved successfully
< Previous
Next >
Finish
Cancel
Capacity
Free
Used S
100 GB
99,9 GB
100 GB
99,9 GB
2 repositories
Logged in: .\loic
Connected to: localhost
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History
Type in an object name to search for
Name
Type
Host
Path
Capacity
Free
Used S
Default Backup Repository
Windows
WIN-G1FR2FSD...
E:\Backup
100 GB
99,9 GB
Sauvegarde
Windows
WIN-G1FR2FSD...
E:\Sauvegarde
100 GB
99,9 GB
2 repositories
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

ÉTAPE 3 : Ajouter le poste PC1 et déployer l'agent

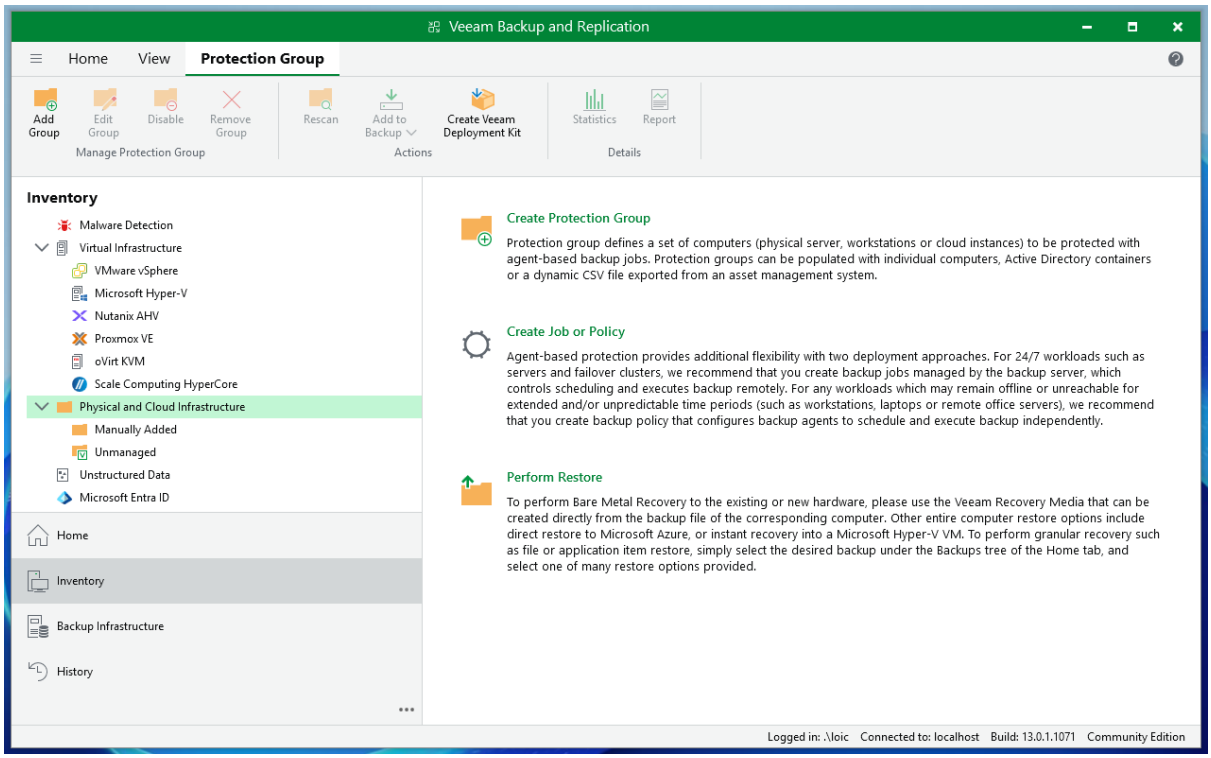
Installation de Windows 11 Pro sur PC1.

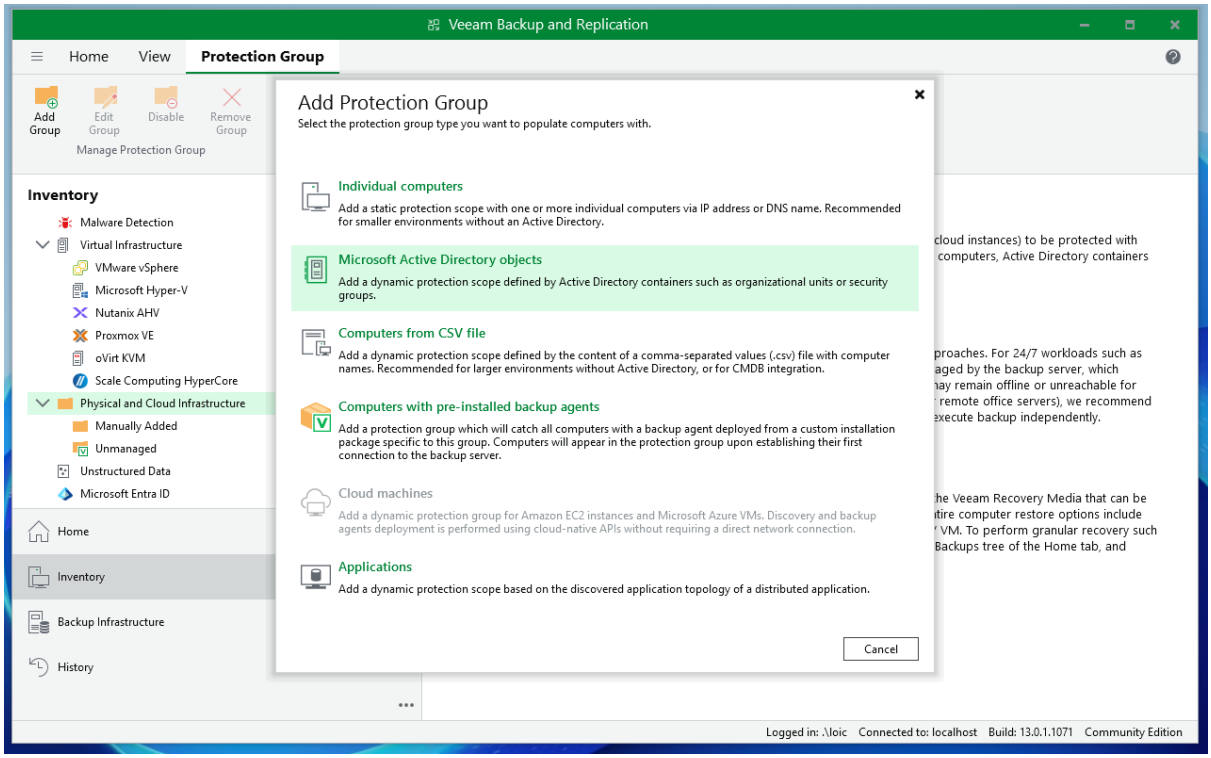




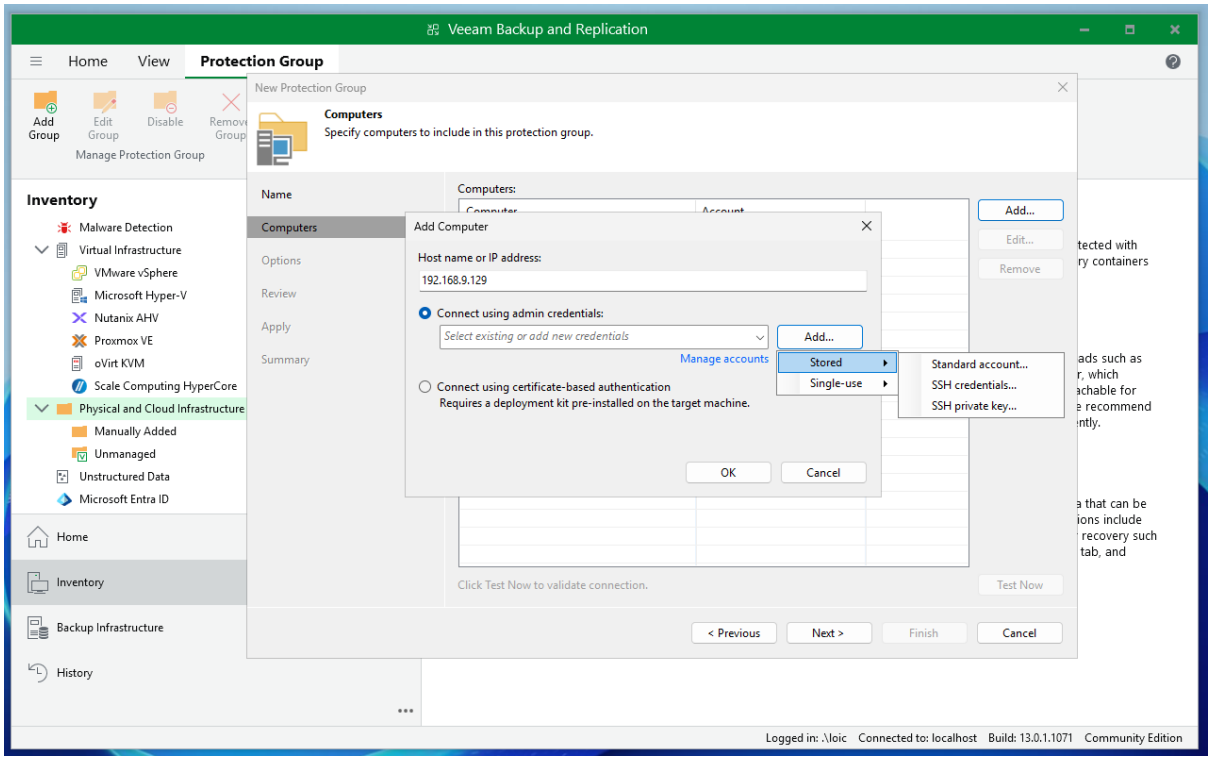
On active le compte Administrateur sur PC1, on définit un mot de passe, et on s'y connecte une première fois.

Puis il faut se rendre dans *Inventory* > *Physical Infrastructure* pour créer un groupe de protection (nommé pc1), et ajouter l'adresse IP de PC1 (ici c'est 192.168.9.129).

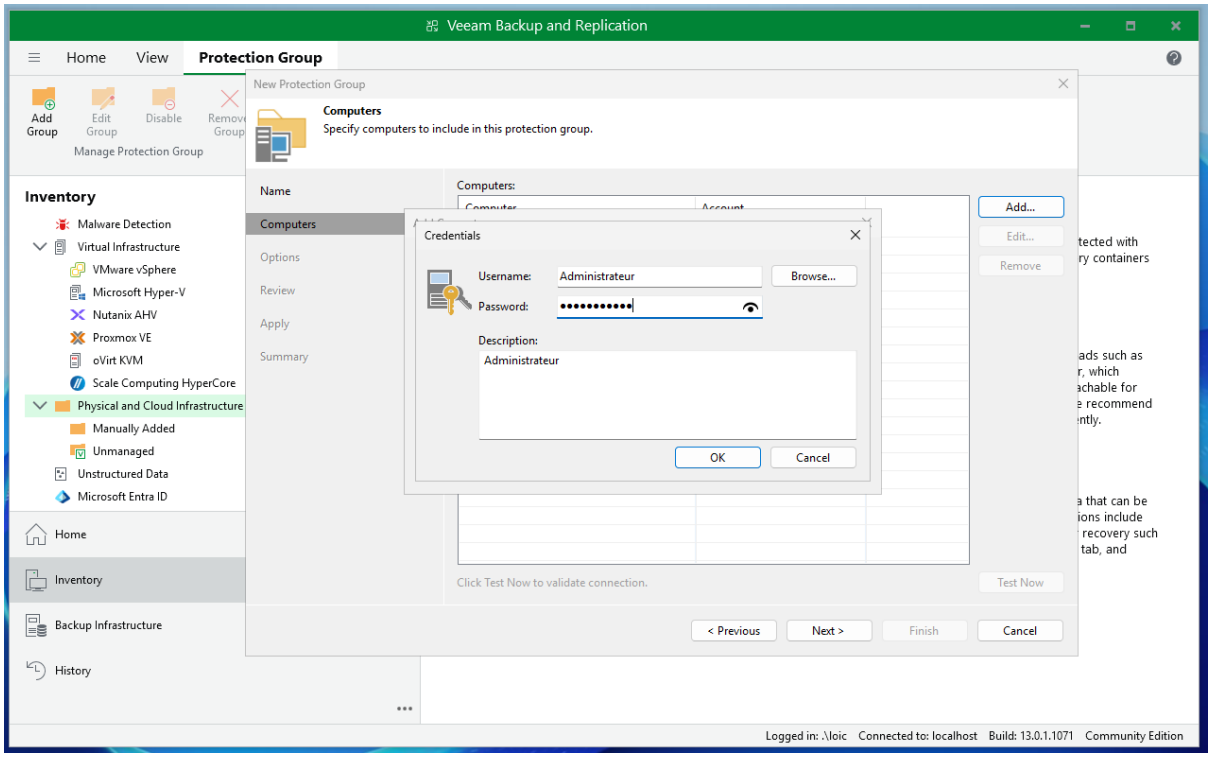


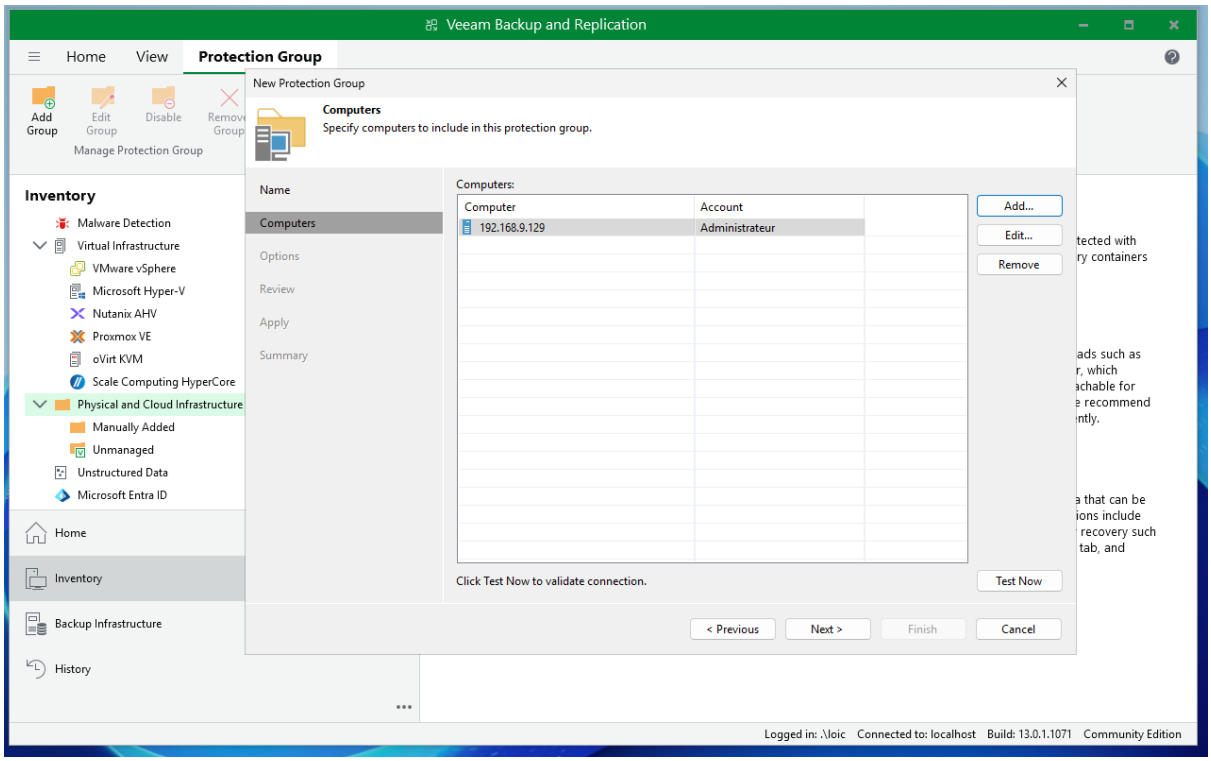


On choisit Individual computers

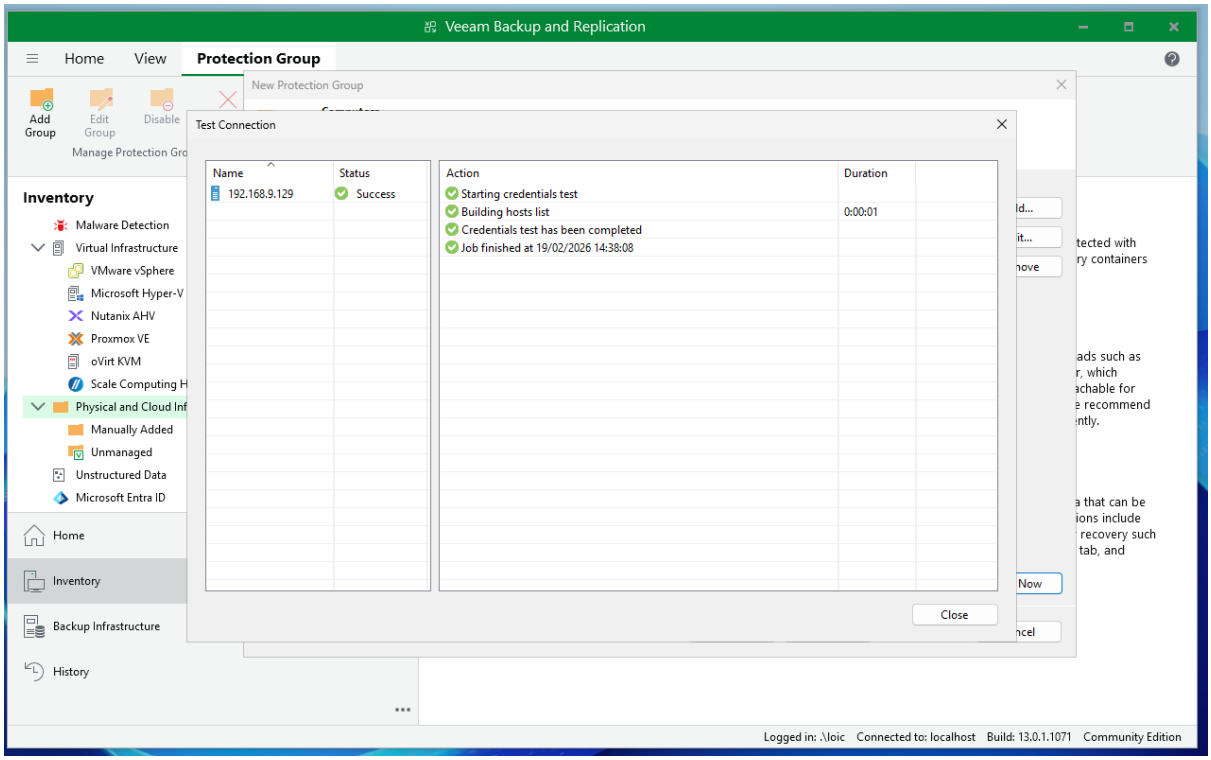


On renseigne le compte administrateur local de PC1 (ex : PC1\Administrateur).

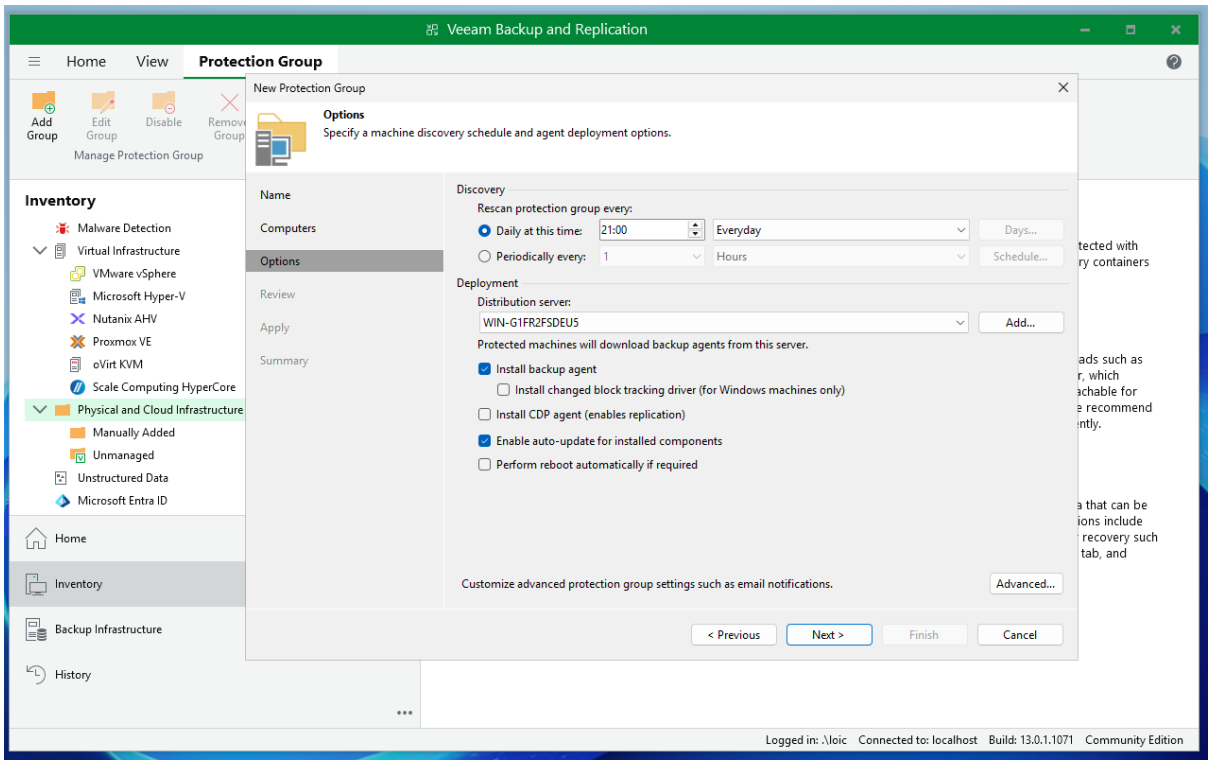


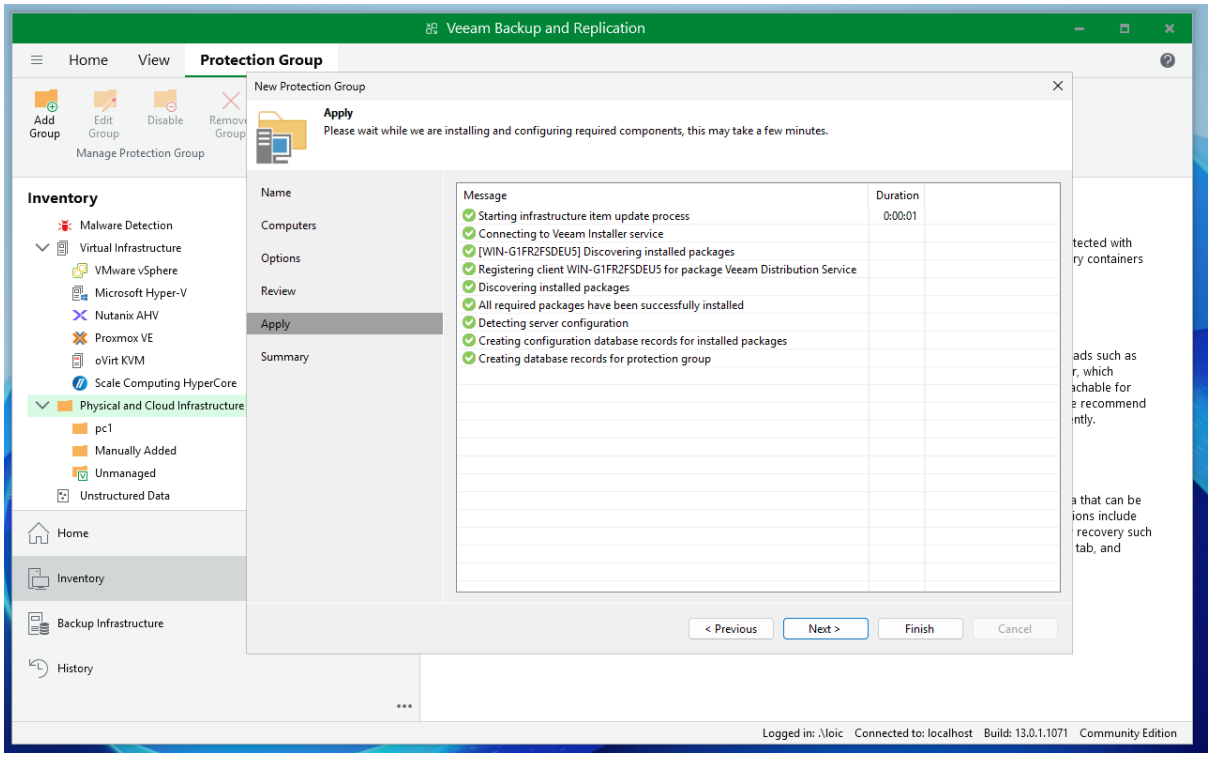


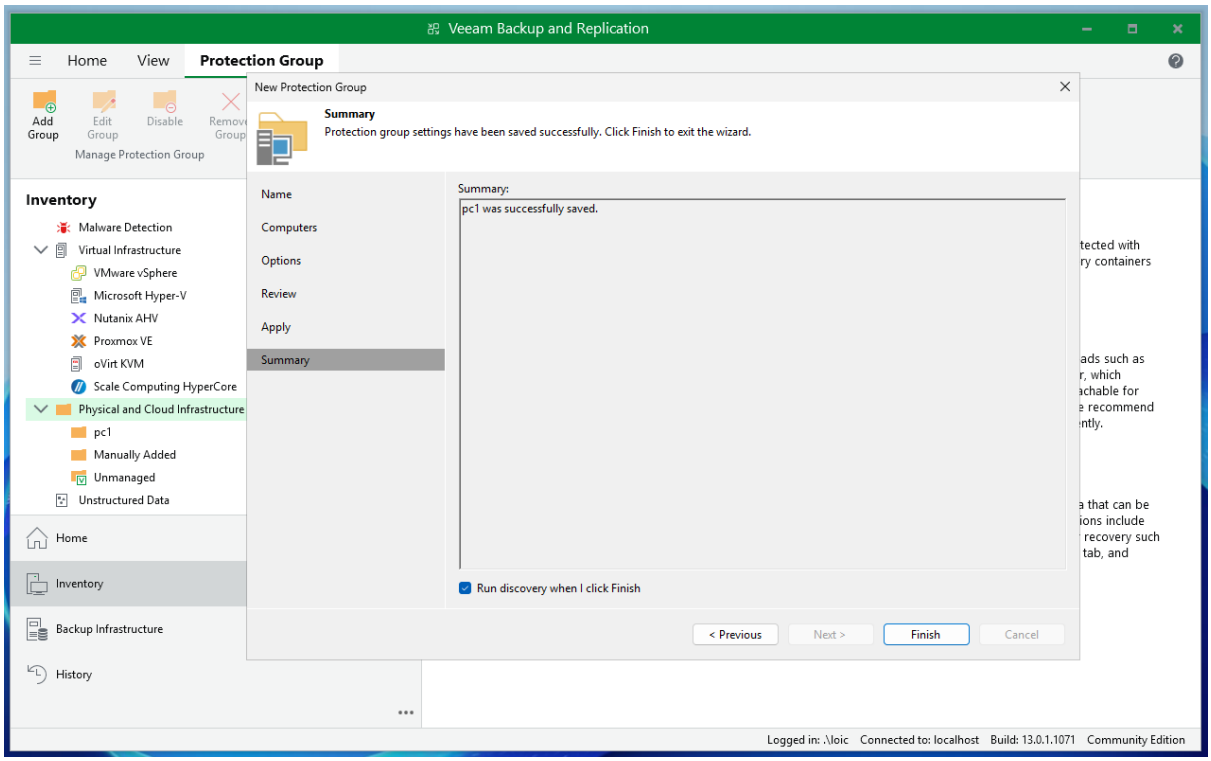
On lance le test de connexion (WMI et RPC doivent réussir). Il faut désactiver le pare feu dans le PC client.

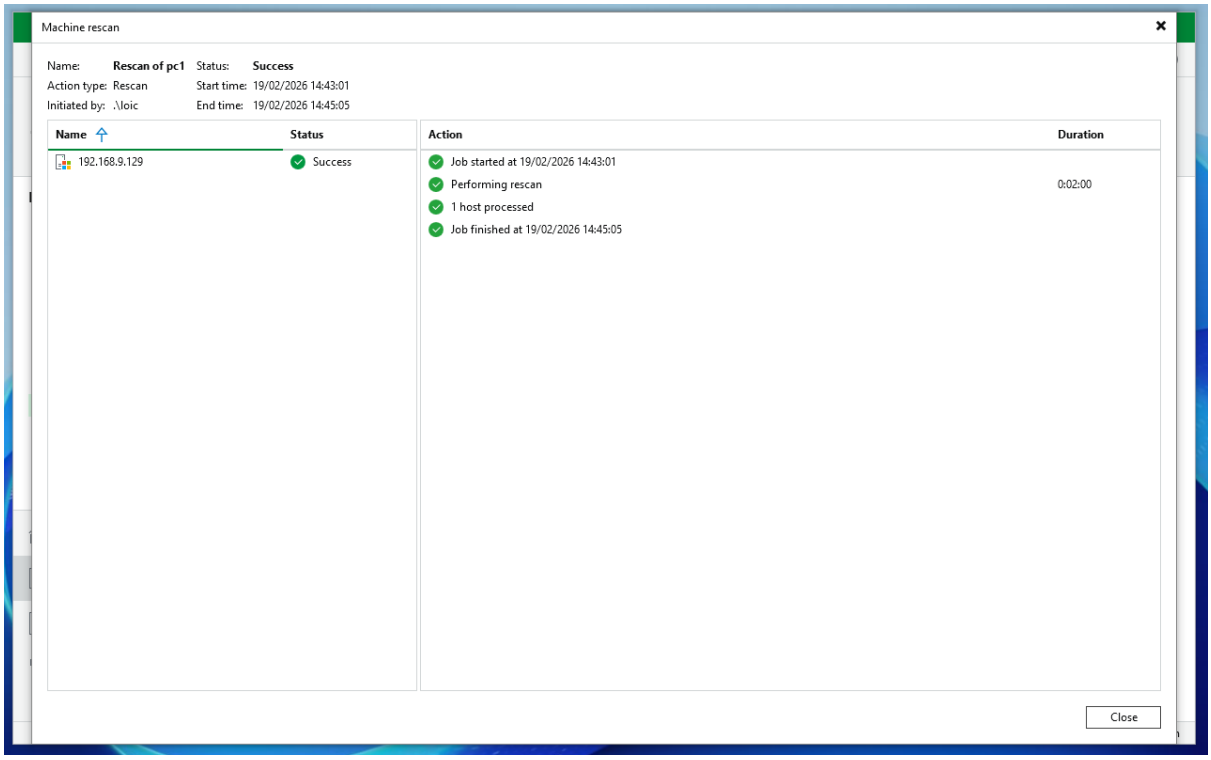


Enfin on valide pour déployer automatiquement l'agent Veeam sur PC1.



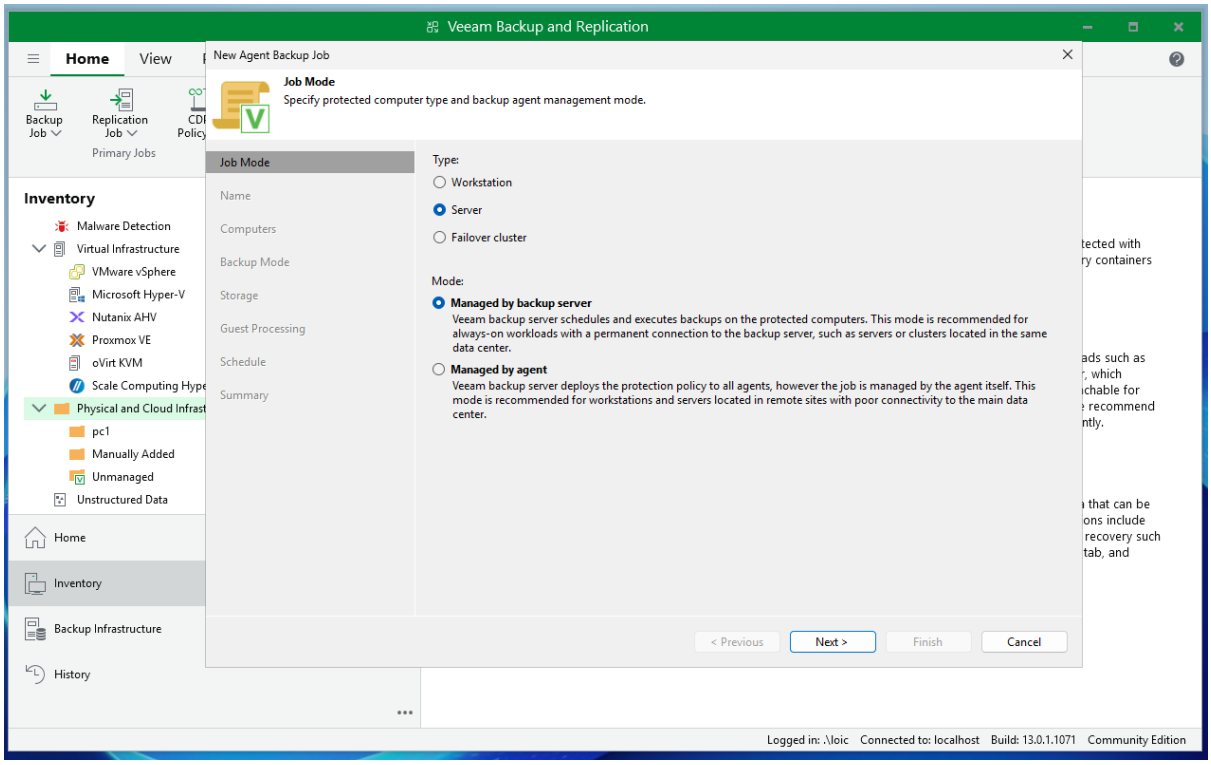

Veeam Backup and Replication
Home
View
Protection Group
Add Group
Edit Group
Disable
Remove Group
Manage Protection Group
New Protection Group
Apply
Please wait while we are installing and configuring required components, this may take a few minutes.
Name
Computers
Options
Review
Apply
Summary
Message
Duration
Starting infrastructure item update process
0:00:01
Connecting to Veeam Installer service
[WIN-G1FR2FSDEU5] Discovering installed packages
Registering client WIN-G1FR2FSDEU5 for package Veeam Distribution Service
Discovering installed packages
All required packages have been successfully installed
Detecting server configuration
Creating configuration database records for installed packages
Creating database records for protection group
< Previous
Next >
Finish
Cancel
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing HyperCore
Physical and Cloud Infrastructure
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition


Veeam Backup and Replication
Home
View
Protection Group
Add Group
Edit Group
Disable
Remove Group
Manage Protection Group
New Protection Group
Summary
Protection group settings have been saved successfully. Click Finish to exit the wizard.
Name
Computers
Options
Review
Apply
Summary
Summary:
pc1 was successfully saved.
Run discovery when I click Finish
< Previous
Next >
Finish
Cancel
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing HyperCore
Physical and Cloud Infrastructure
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

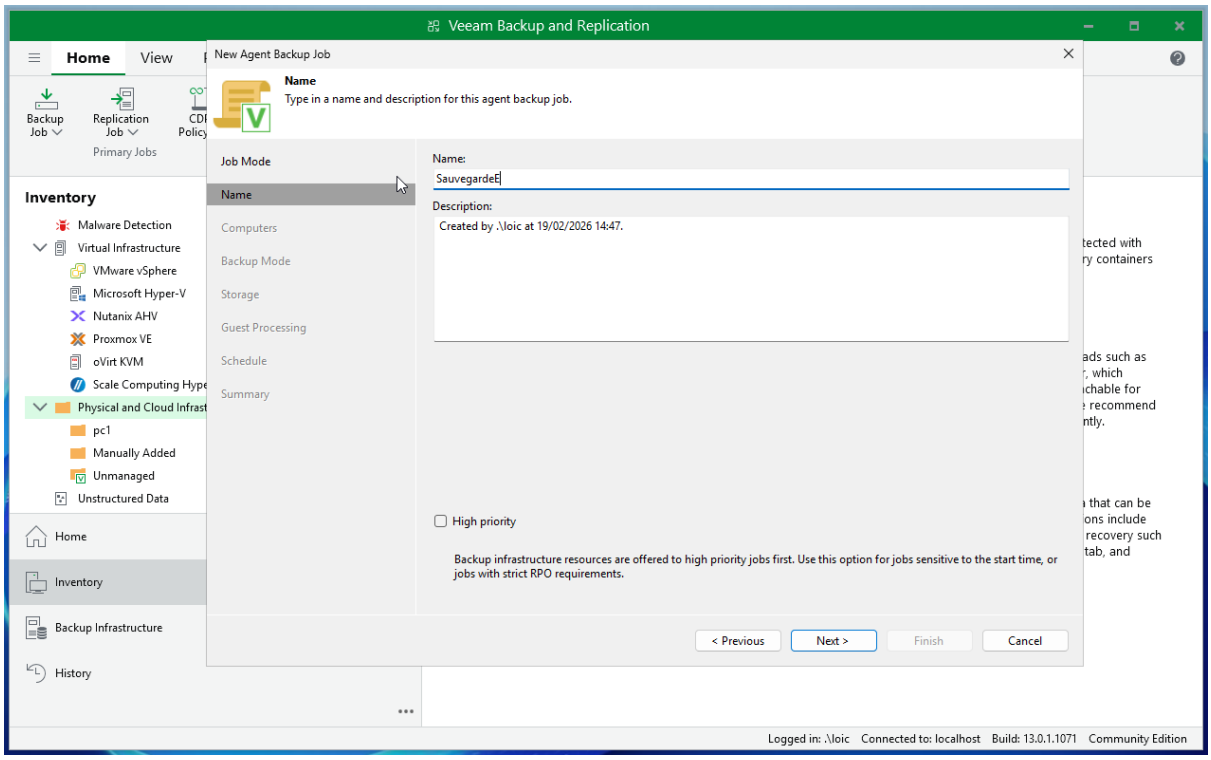

Machine rescan
Name: Rescan of pc1
Status: Success
Action type: Rescan
Start time: 19/02/2026 14:43:01
Initiated by: .\loic
End time: 19/02/2026 14:45:05
Name
Status
Action
Duration
192.168.9.129
Success
Job started at 19/02/2026 14:43:01
Performing rescan
0:02:00
1 host processed
Job finished at 19/02/2026 14:45:05
Close

ÉTAPE 4 : Créer une tâche de sauvegarde

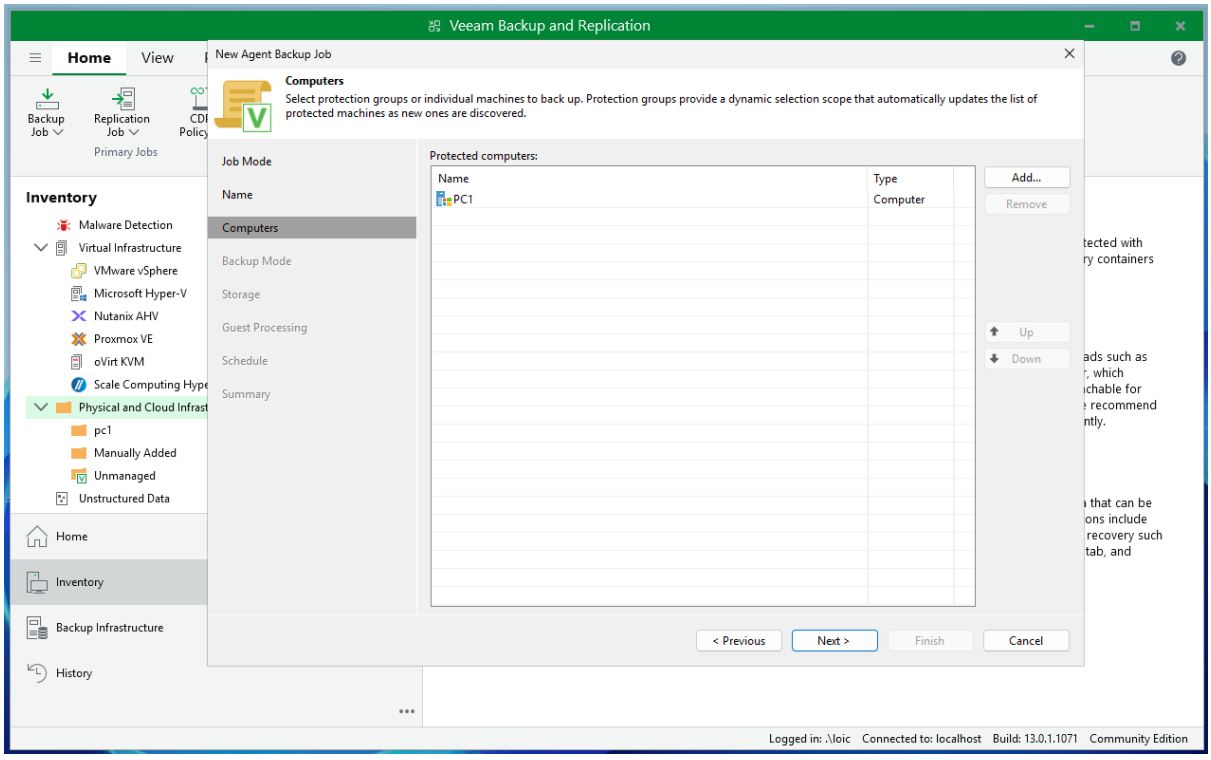
Dans *Home*, on clique sur *Backup Job* > *Windows Computer*.

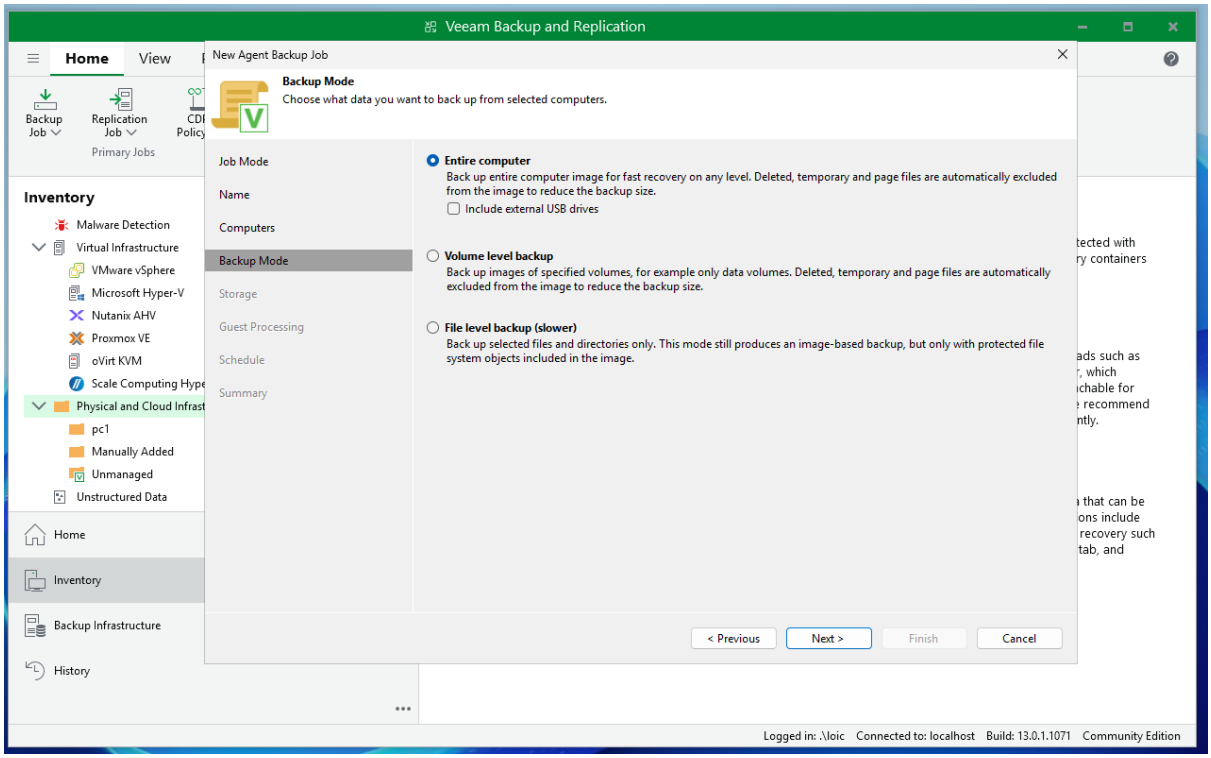


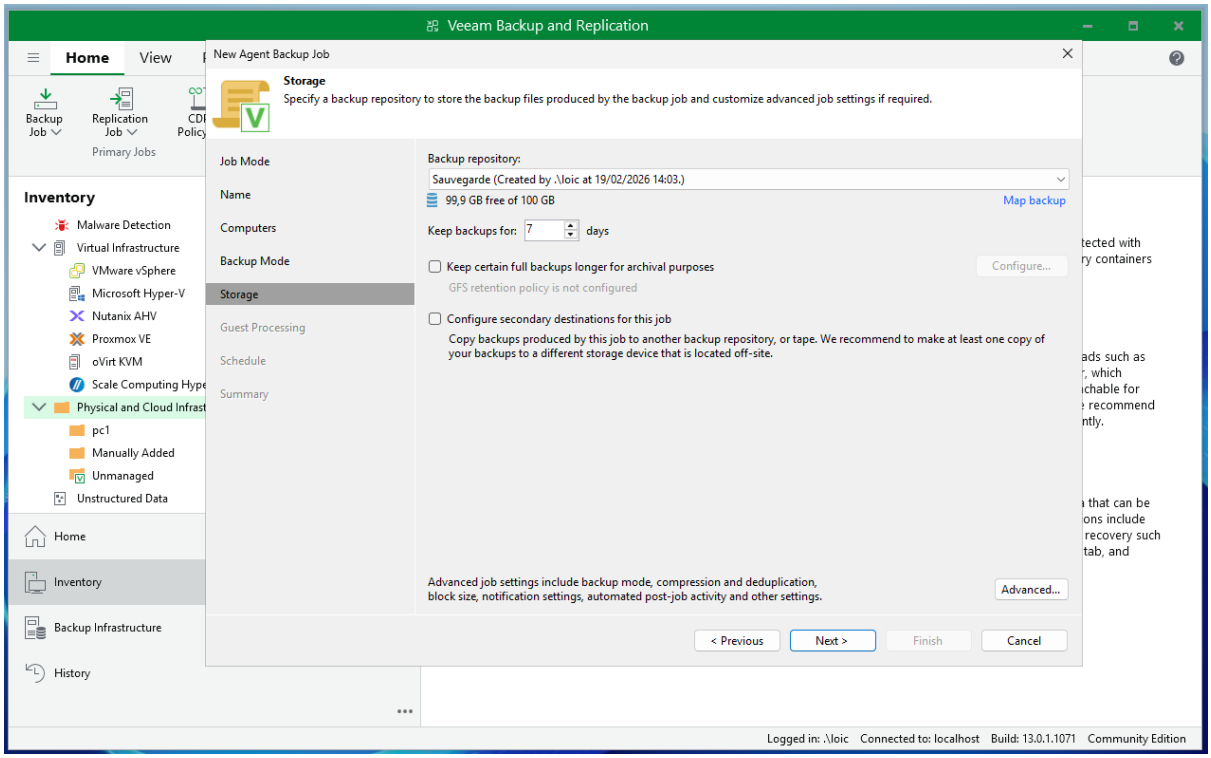
On nomme la tâche *SauvegardeE*.

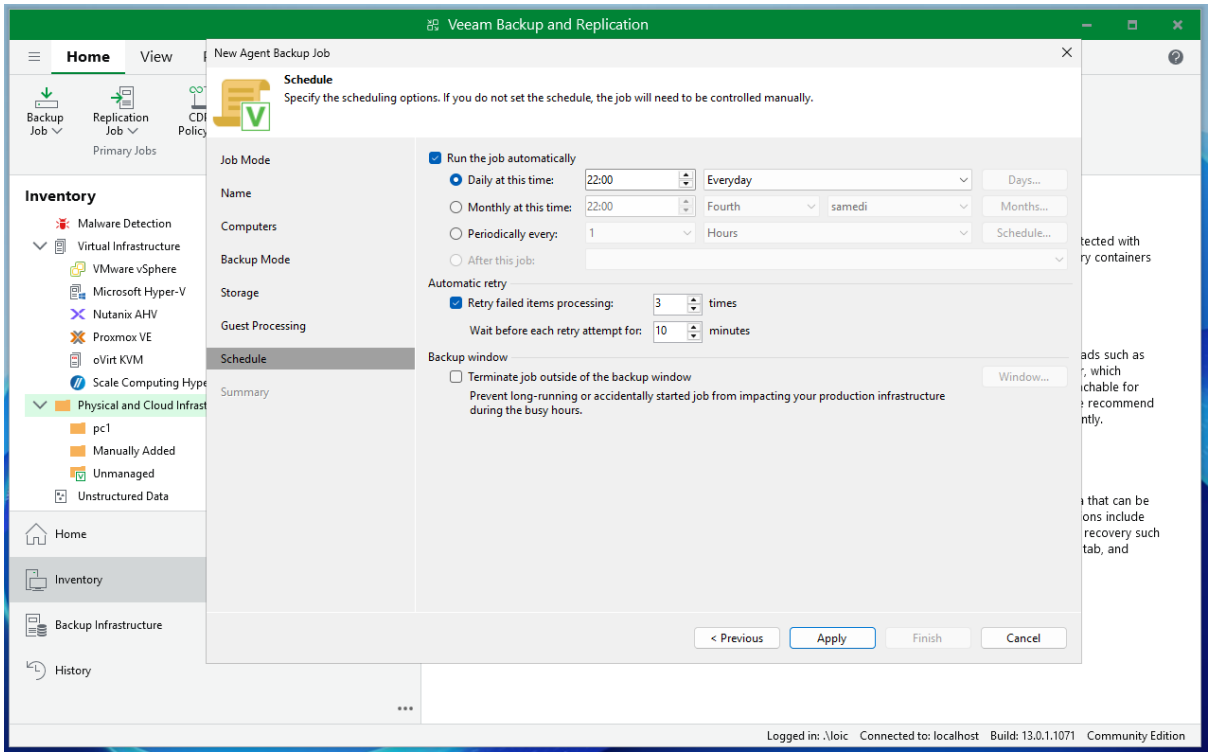


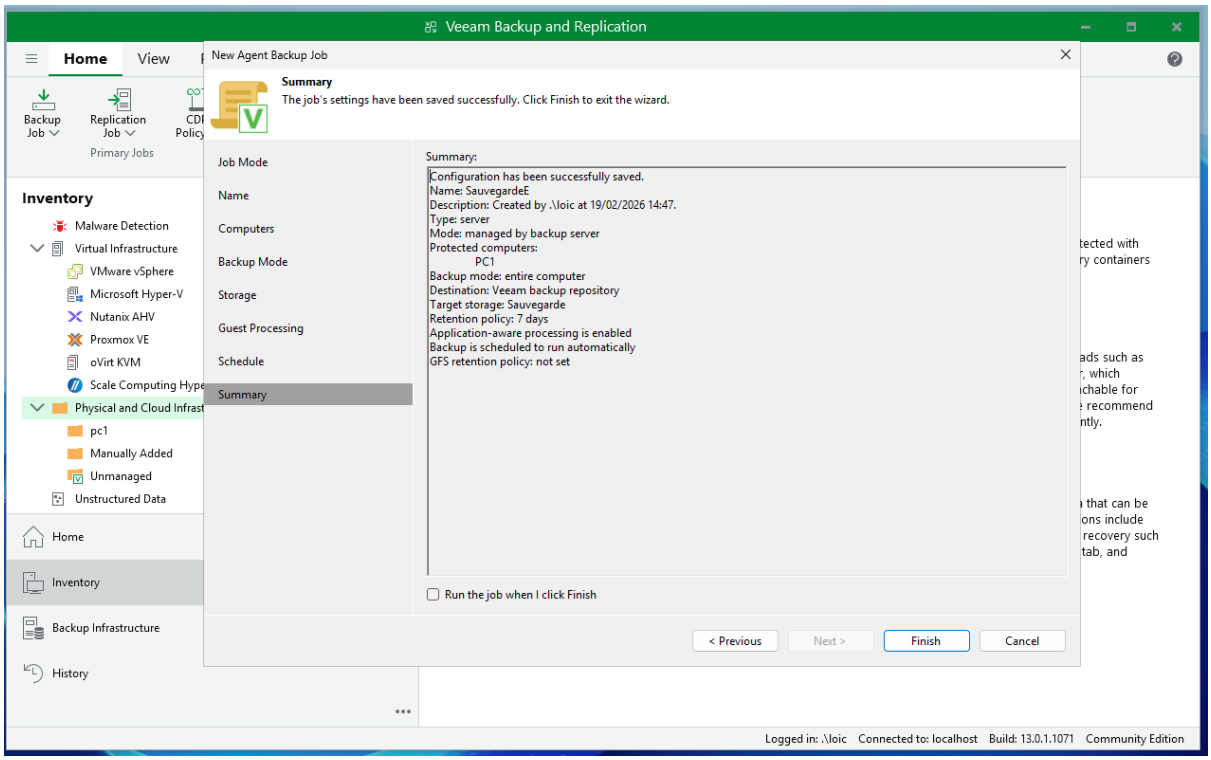
Il faut définir comme cible le dépôt *Sauvegarde*, et planifier l'exécution chaque jour à 22h, avec une rétention de 7 jours.

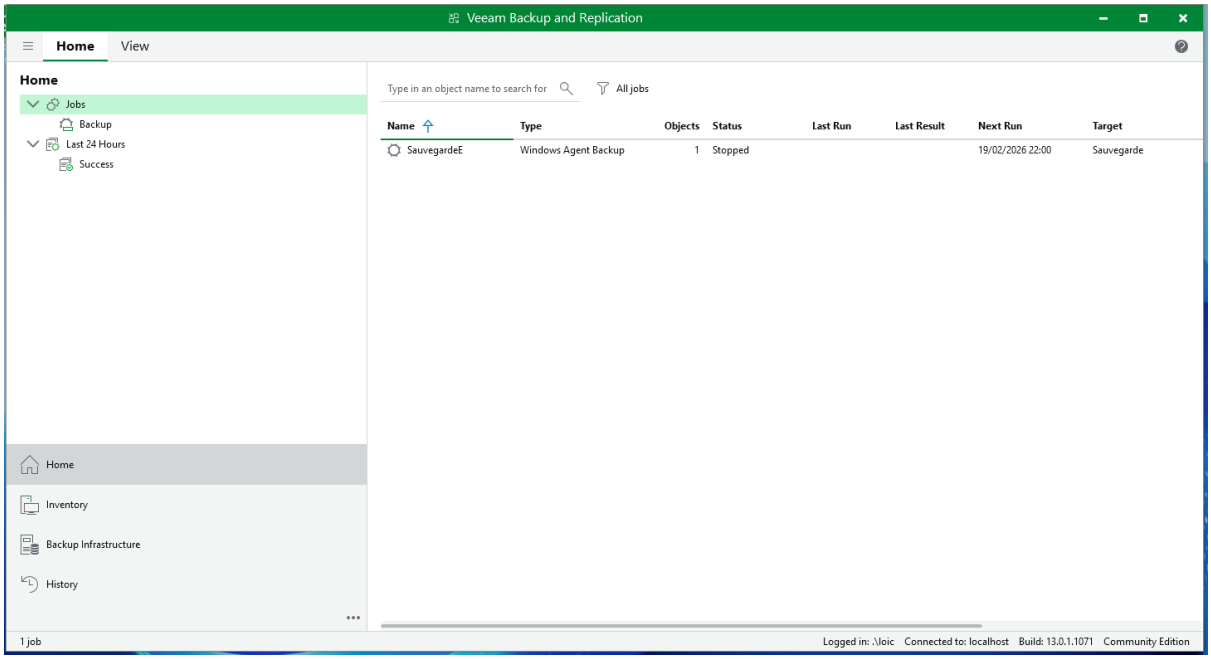



Veeam Backup and Replication
Home
View
Backup Job
Replication Job
Primary Jobs
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing Hype
Physical and Cloud Infrast
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
New Agent Backup Job
Backup Mode
Choose what data you want to back up from selected computers.
Job Mode
Name
Computers
Backup Mode
Storage
Guest Processing
Schedule
Summary
Entire computer
Back up entire computer image for fast recovery on any level. Deleted, temporary and page files are automatically excluded from the image to reduce the backup size.
Include external USB drives
Volume level backup
Back up images of specified volumes, for example only data volumes. Deleted, temporary and page files are automatically excluded from the image to reduce the backup size.
File level backup (slower)
Back up selected files and directories only. This mode still produces an image-based backup, but only with protected file system objects included in the image.
< Previous
Next >
Finish
Cancel
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition


Veeam Backup and Replication
Home
View
Backup Job
Replication Job
Primary Jobs
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing Hype
Physical and Cloud Infrast
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
New Agent Backup Job
Storage
Specify a backup repository to store the backup files produced by the backup job and customize advanced job settings if required.
Job Mode
Name
Computers
Backup Mode
Storage
Guest Processing
Schedule
Summary
Backup repository:
Sauvegarde (Created by .\loic at 19/02/2026 14:03.)
99,9 GB free of 100 GB
Map backup
Keep backups for: 7 days
Keep certain full backups longer for archival purposes
Configure...
GFS retention policy is not configured
Configure secondary destinations for this job
Copy backups produced by this job to another backup repository, or tape. We recommend to make at least one copy of your backups to a different storage device that is located off-site.
Advanced job settings include backup mode, compression and deduplication, block size, notification settings, automated post-job activity and other settings.
Advanced...
< Previous
Next >
Finish
Cancel
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

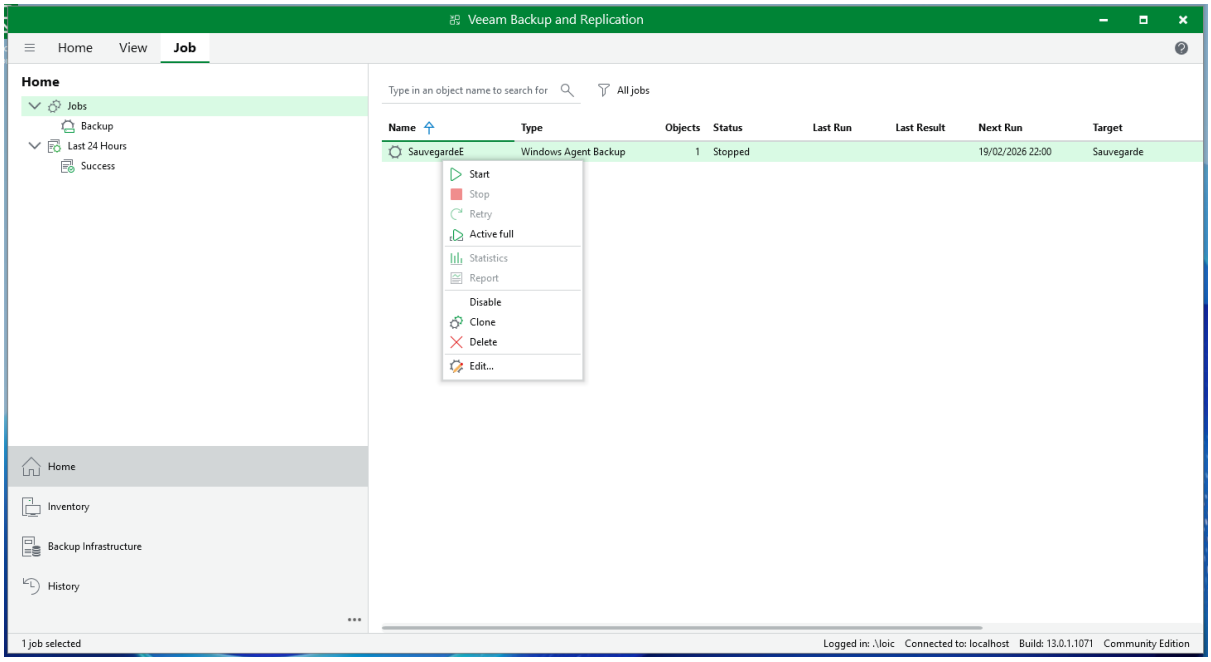

Veeam Backup and Replication
Home
View
Backup Job
Replication Job
Primary Jobs
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing Hype
Physical and Cloud Infrast
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
New Agent Backup Job
Schedule
Specify the scheduling options. If you do not set the schedule, the job will need to be controlled manually.
Job Mode
Name
Computers
Backup Mode
Storage
Guest Processing
Schedule
Summary
Run the job automatically
Daily at this time:
22:00
Everyday
Days...
Monthly at this time:
22:00
Fourth
samedi
Months...
Periodically every:
1
Hours
Schedule...
After this job:
Automatic retry
Retry failed items processing:
3
times
Wait before each retry attempt for:
10
minutes
Backup window
Terminate job outside of the backup window
Window...
Prevent long-running or accidentally started job from impacting your production infrastructure during the busy hours.
< Previous
Apply
Finish
Cancel
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

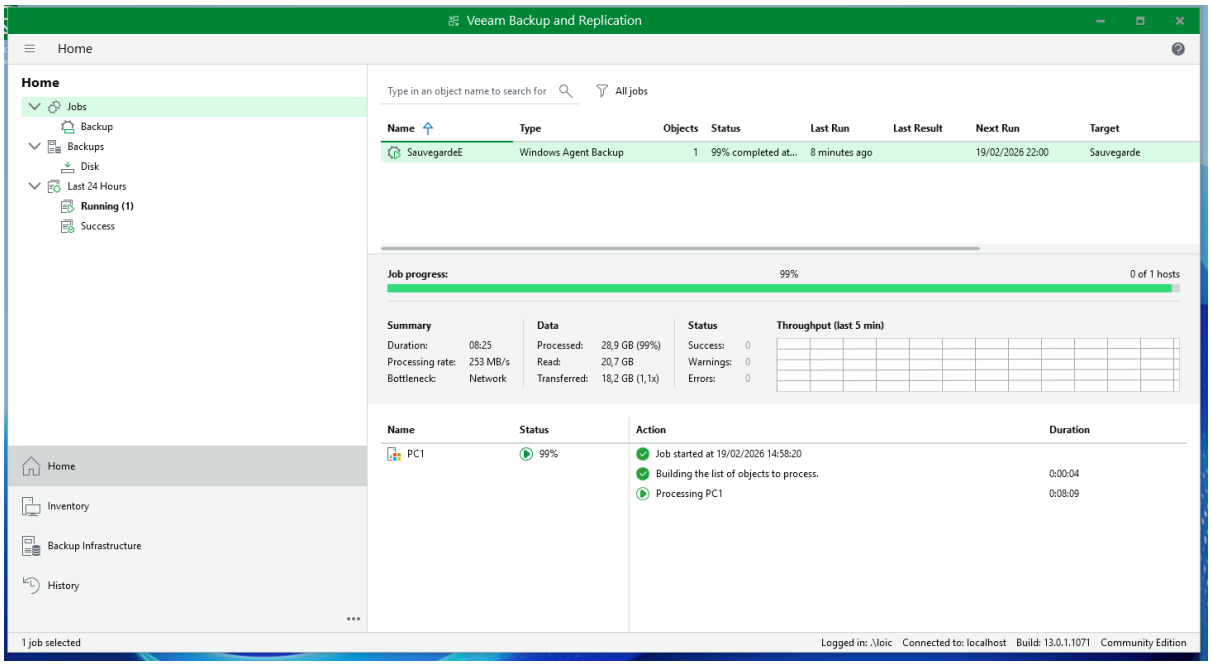

Veeam Backup and Replication
Home
View
Backup Job
Replication Job
Primary Jobs
Inventory
Malware Detection
Virtual Infrastructure
VMware vSphere
Microsoft Hyper-V
Nutanix AHV
Proxmox VE
oVirt KVM
Scale Computing Hype
Physical and Cloud Infrast
pc1
Manually Added
Unmanaged
Unstructured Data
Home
Inventory
Backup Infrastructure
History
New Agent Backup Job
Summary
The job's settings have been saved successfully. Click Finish to exit the wizard.
Job Mode
Name
Computers
Backup Mode
Storage
Guest Processing
Schedule
Summary
Summary:
Configuration has been successfully saved.
Name: SauvegardeE
Description: Created by .\loic at 19/02/2026 14:47.
Type: server
Mode: managed by backup server
Protected computers:
PC1
Backup mode: entire computer
Destination: Veeam backup repository
Target storage: Sauvegarde
Retention policy: 7 days
Application-aware processing is enabled
Backup is scheduled to run automatically
GFS retention policy: not set
Run the job when I click Finish
< Previous
Next >
Finish
Cancel
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

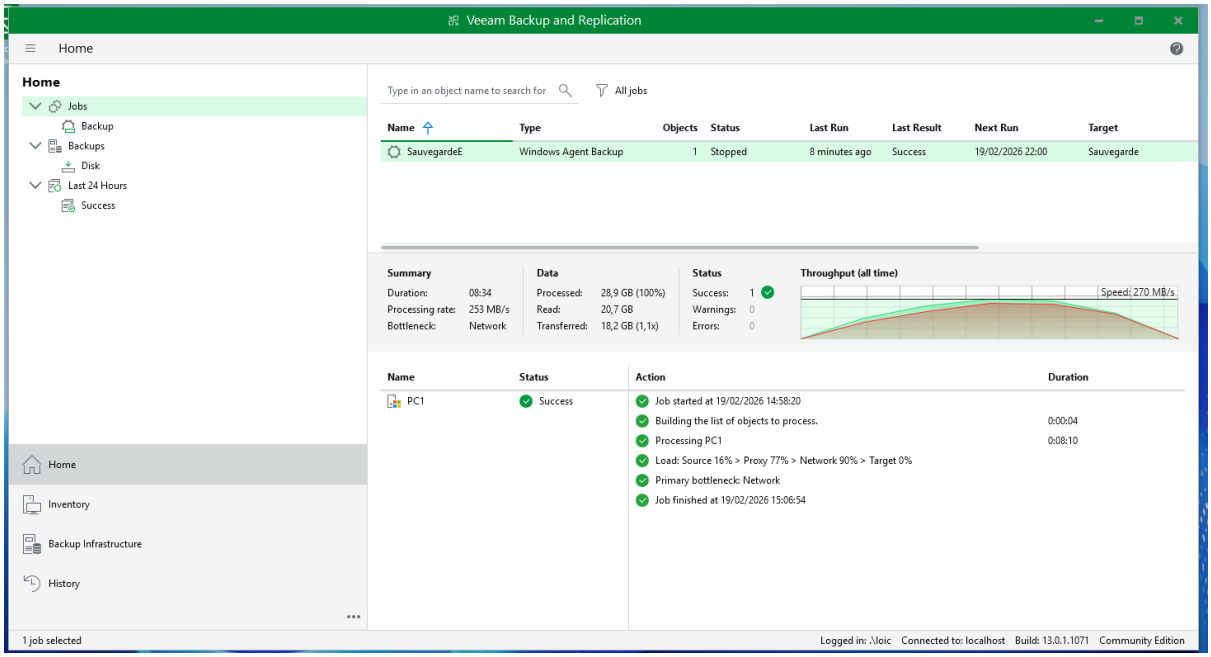

Veeam Backup and Replication
Home
View
Home
Jobs
Backup
Last 24 Hours
Success
Type in an object name to search for
All jobs
Name
Type
Objects
Status
Last Run
Last Result
Next Run
Target
SauvegardeE
Windows Agent Backup
1
Stopped
19/02/2026 22:00
Sauvegarde
Home
Inventory
Backup Infrastructure
History
1 job
Logged in: .\loic
Connected to: localhost
Build: 13.0.1.1071
Community Edition

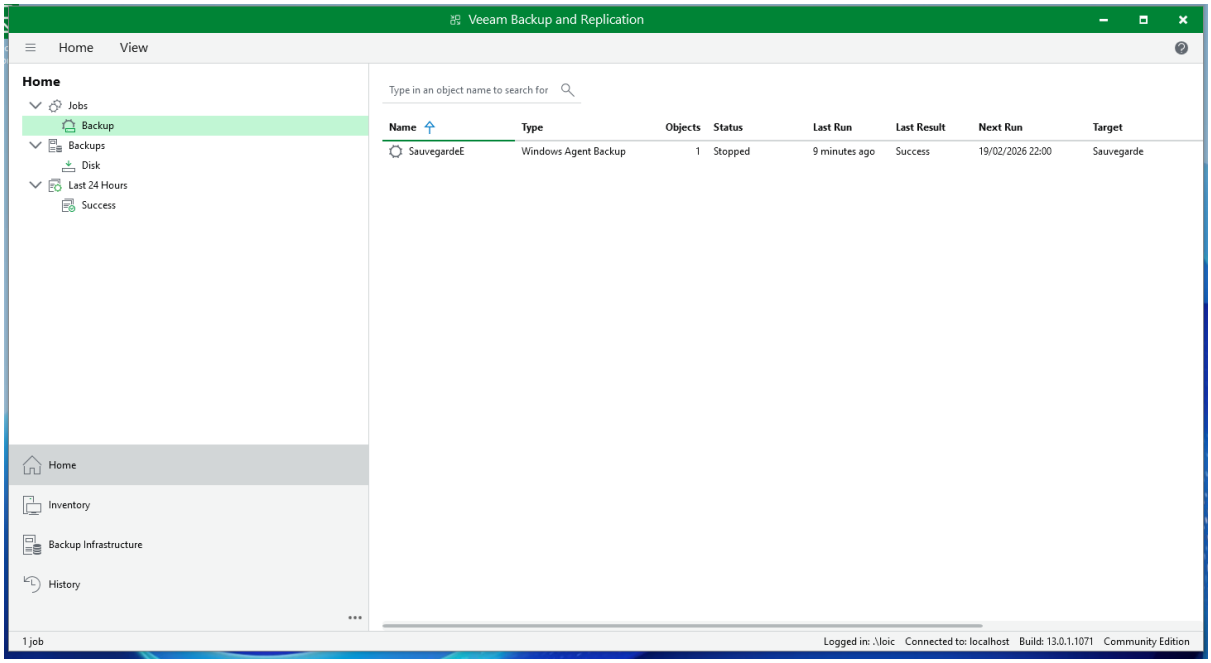
ÉTAPE 5 : Lancer une première sauvegarde

Il faut que l'on sélectionne la tâche *SauvegardeE* (dans *Home* > *Jobs*), et on clique sur *Start* pour exécuter manuellement la sauvegarde.





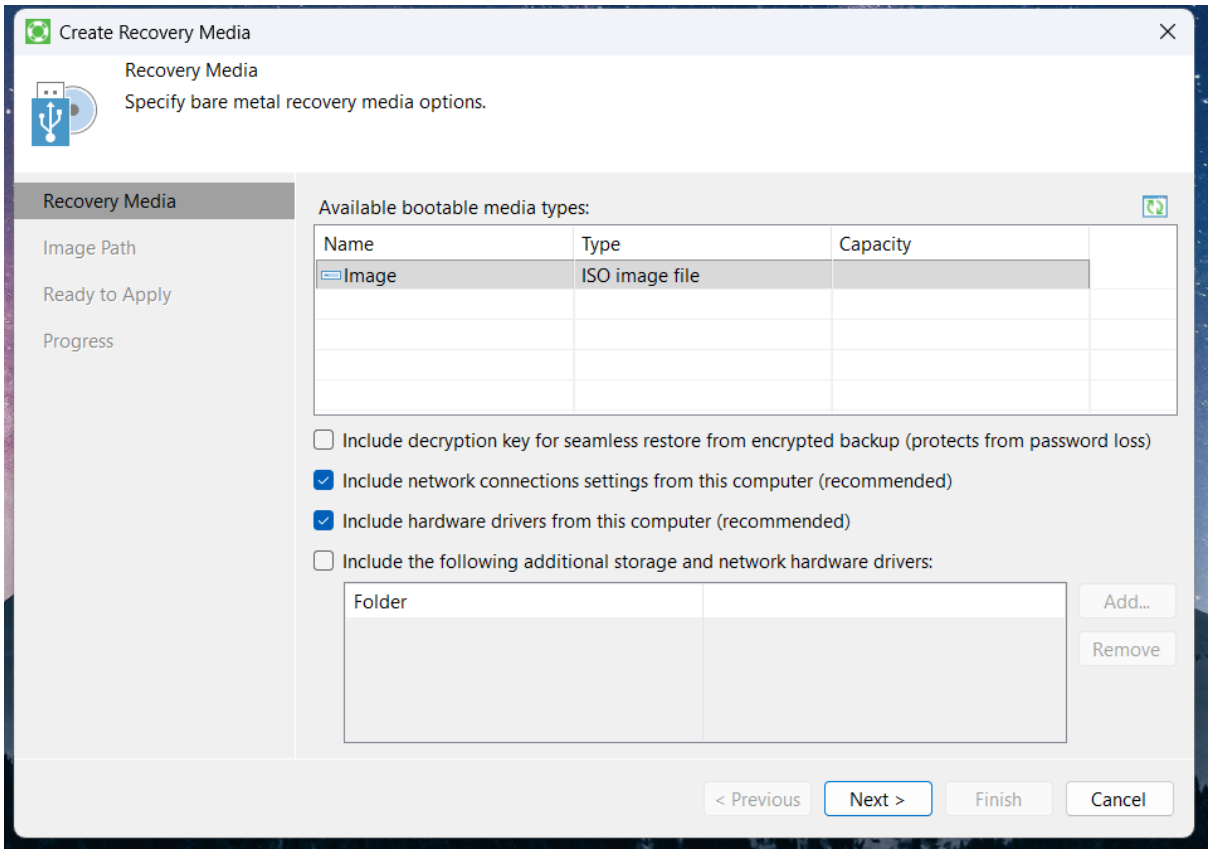


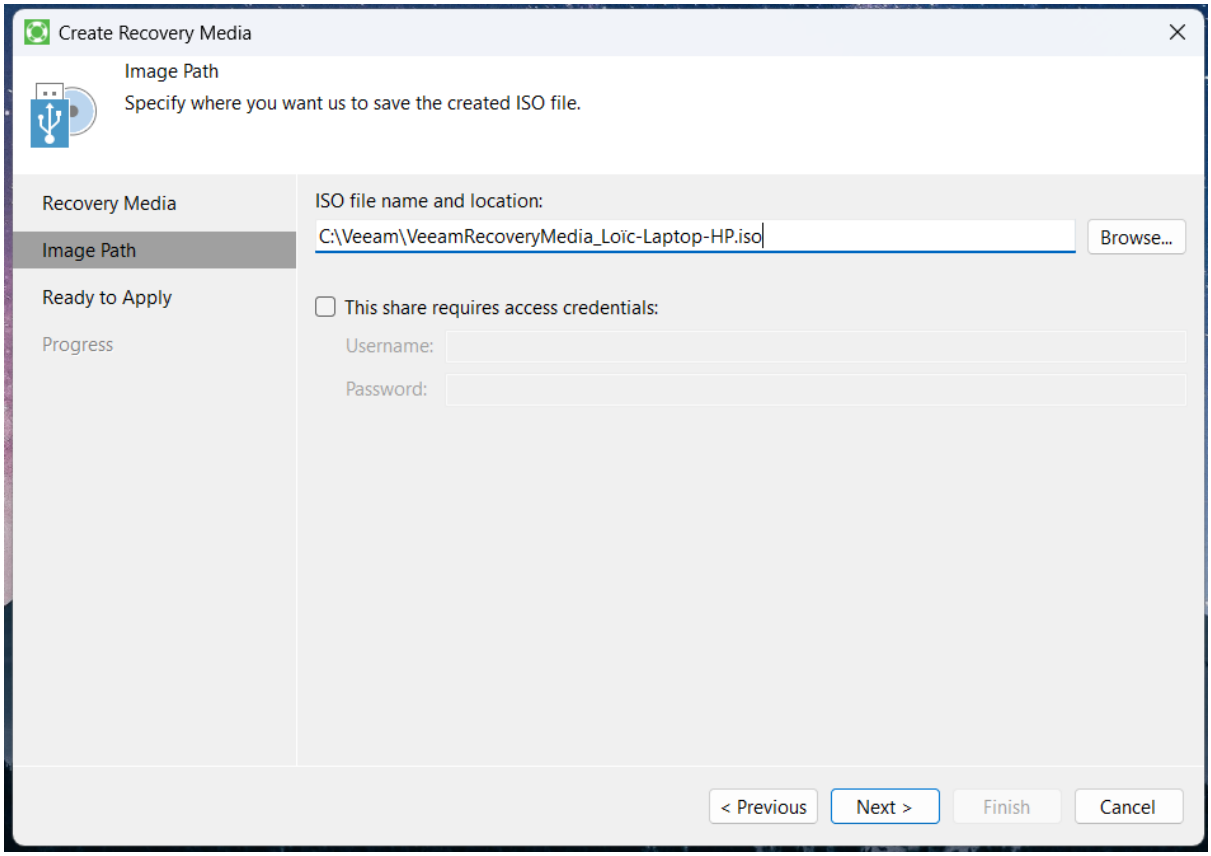


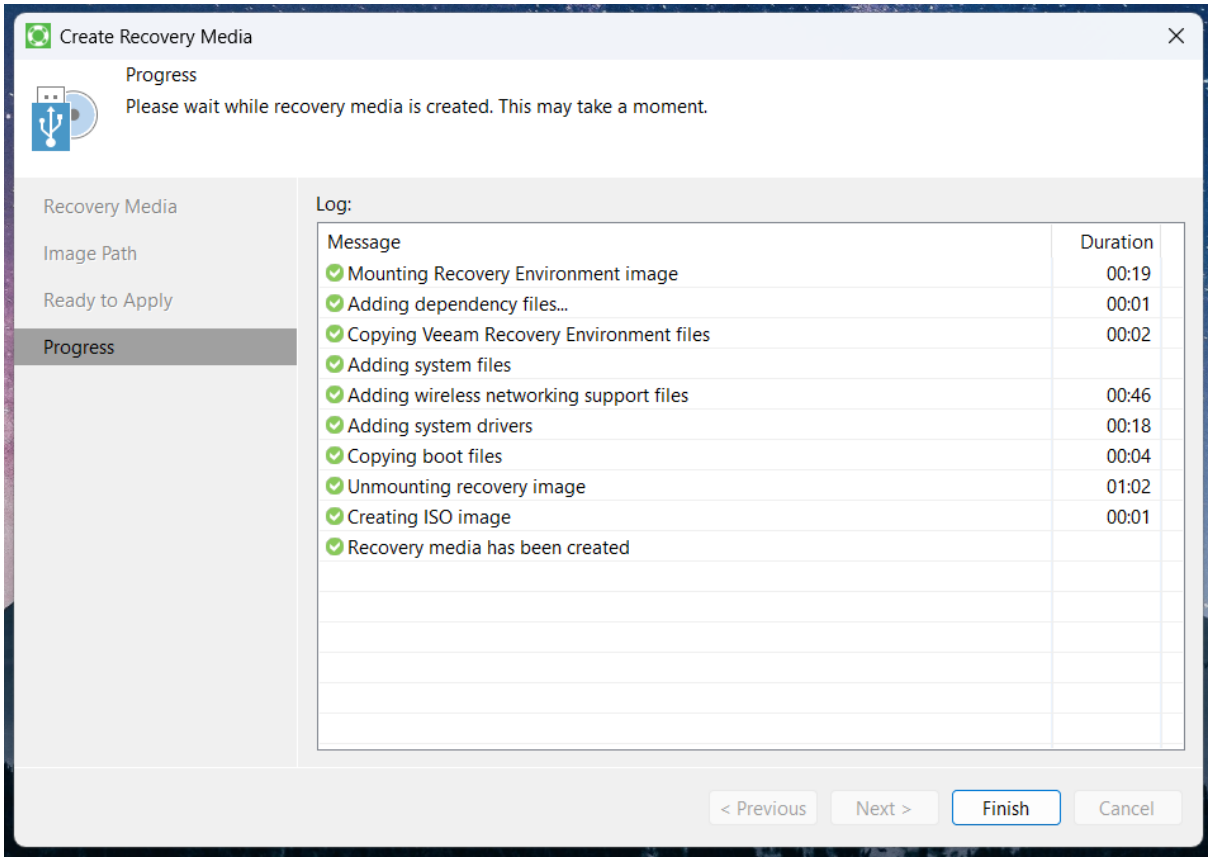
Il faut Vérifier que le fichier .vbk est bien généré dans le dépôt *E:\Sauvegarde*.

ÉTAPE 6 : Créer le support ISO de récupération

Sur la machine physique (sur laquelle tourne la machine virtuelle), il faut installer *Veeam Agent for Microsoft Windows Community Edition*. Cela va permettre de créer l'ISO.

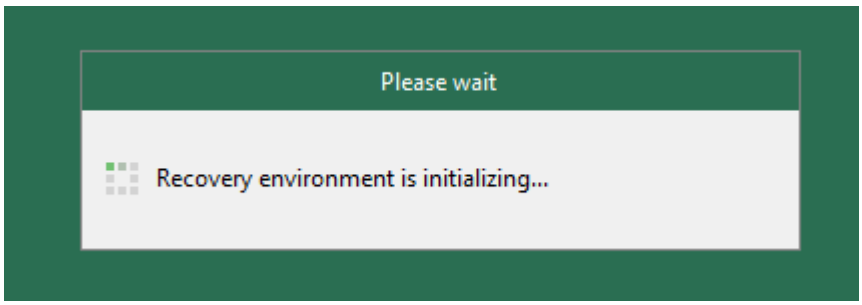



Create Recovery Media
Image Path
Specify where you want us to save the created ISO file.
Recovery Media
Image Path
Ready to Apply
Progress
ISO file name and location:
C:\Veeam\VeeamRecoveryMedia_Loïc-Laptop-HP.iso
Browse...
This share requires access credentials:
Username:
Password:
< Previous
Next >
Finish
Cancel


Create Recovery Media
Progress
Please wait while recovery media is created. This may take a moment.
Recovery Media
Image Path
Ready to Apply
Progress
Log:
Message
Duration
Mounting Recovery Environment image 00:19
Adding dependency files... 00:01
Copying Veeam Recovery Environment files 00:02
Adding system files
Adding wireless networking support files 00:46
Adding system drivers 00:18
Copying boot files 00:04
Unmounting recovery image 01:02
Creating ISO image 00:01
Recovery media has been created
< Previous
Next >
Finish
Cancel

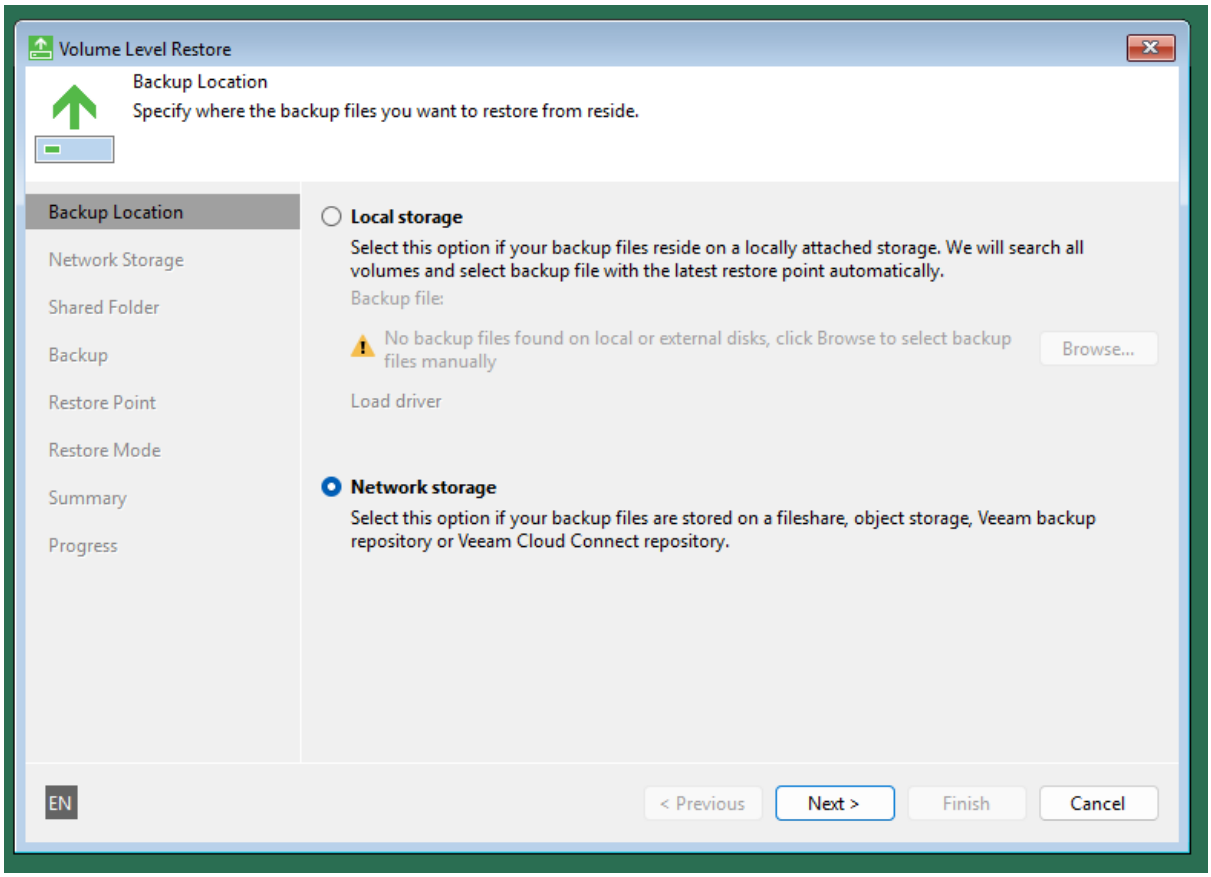
ÉTAPE 7 : Démarrer une restauration complète

On créé une nouvelle machine virtuelle et on injecte l'ISO que nous avons créé précédemment dans le lecteur CD. On choisit *Bare Metal Recovery*.

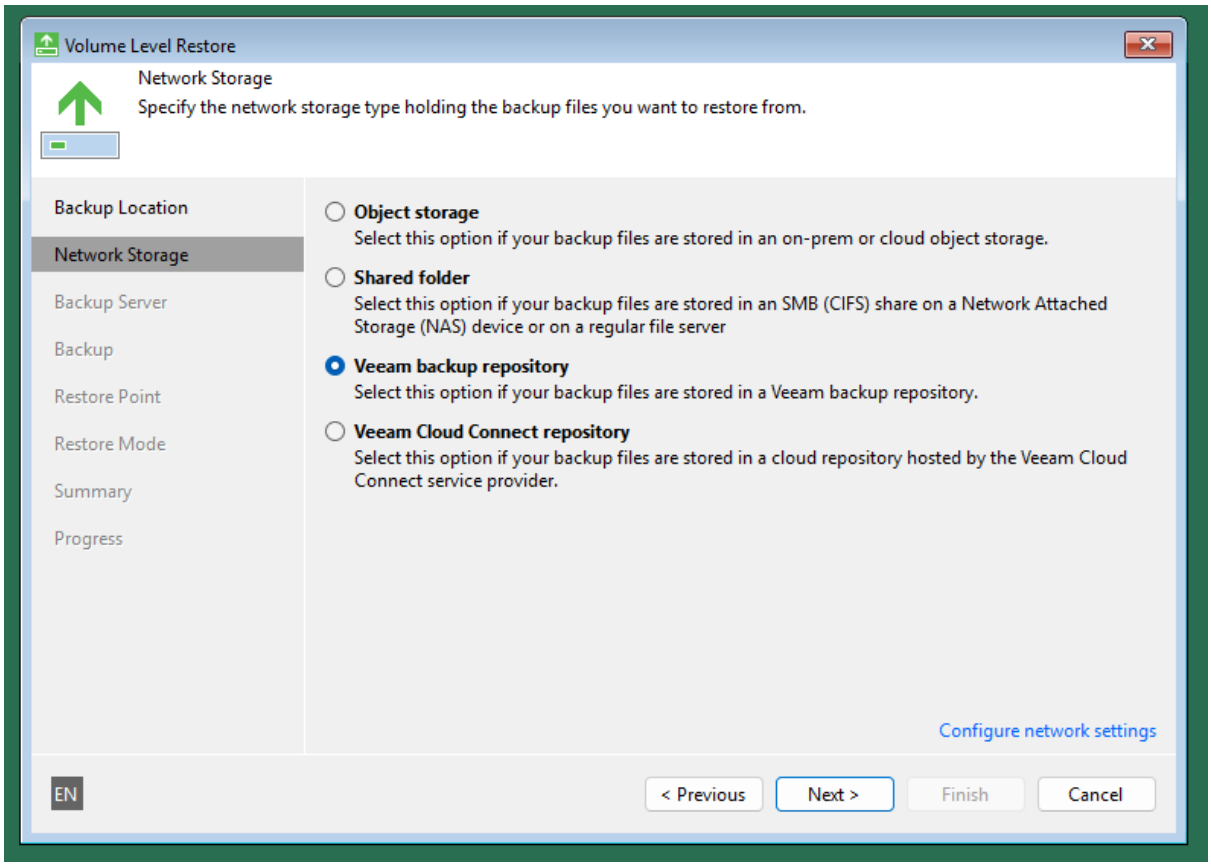




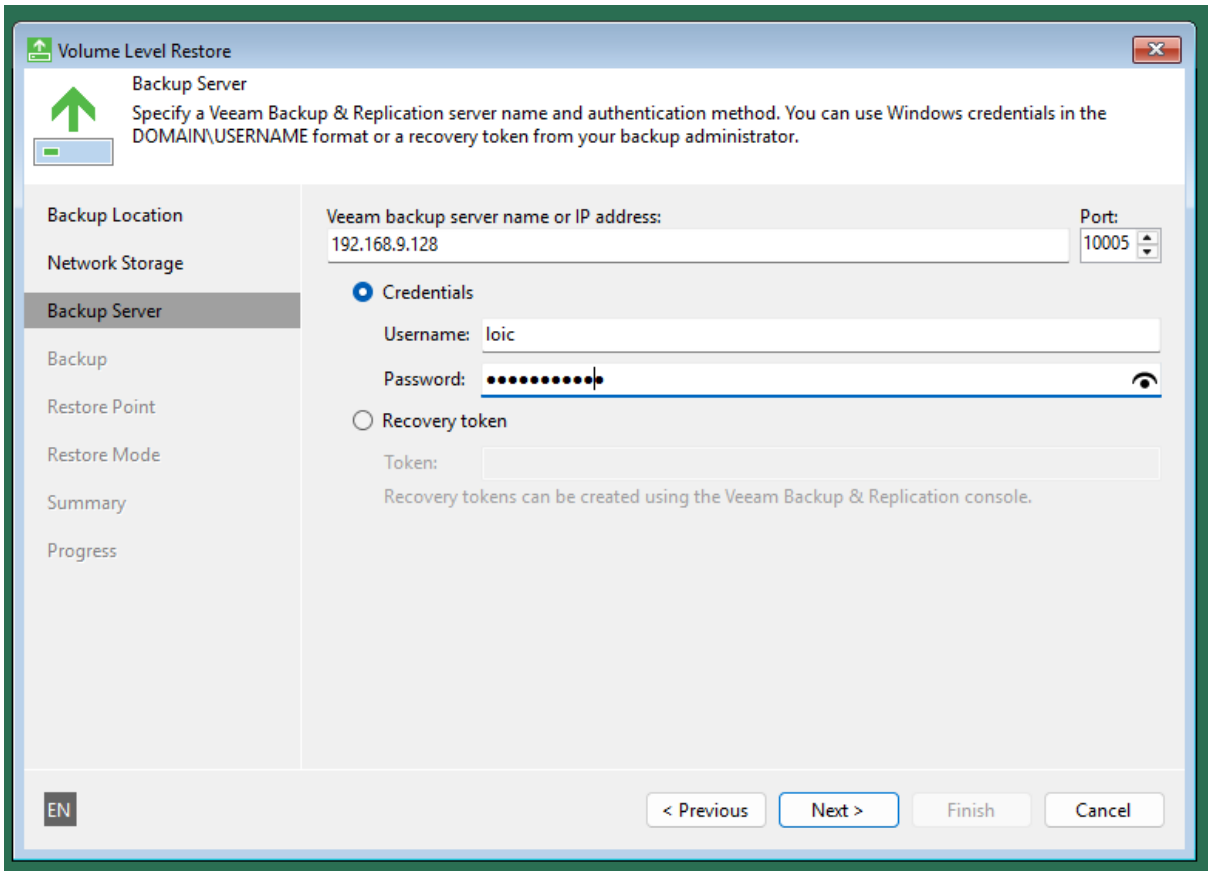
On choisit *Network storage*.

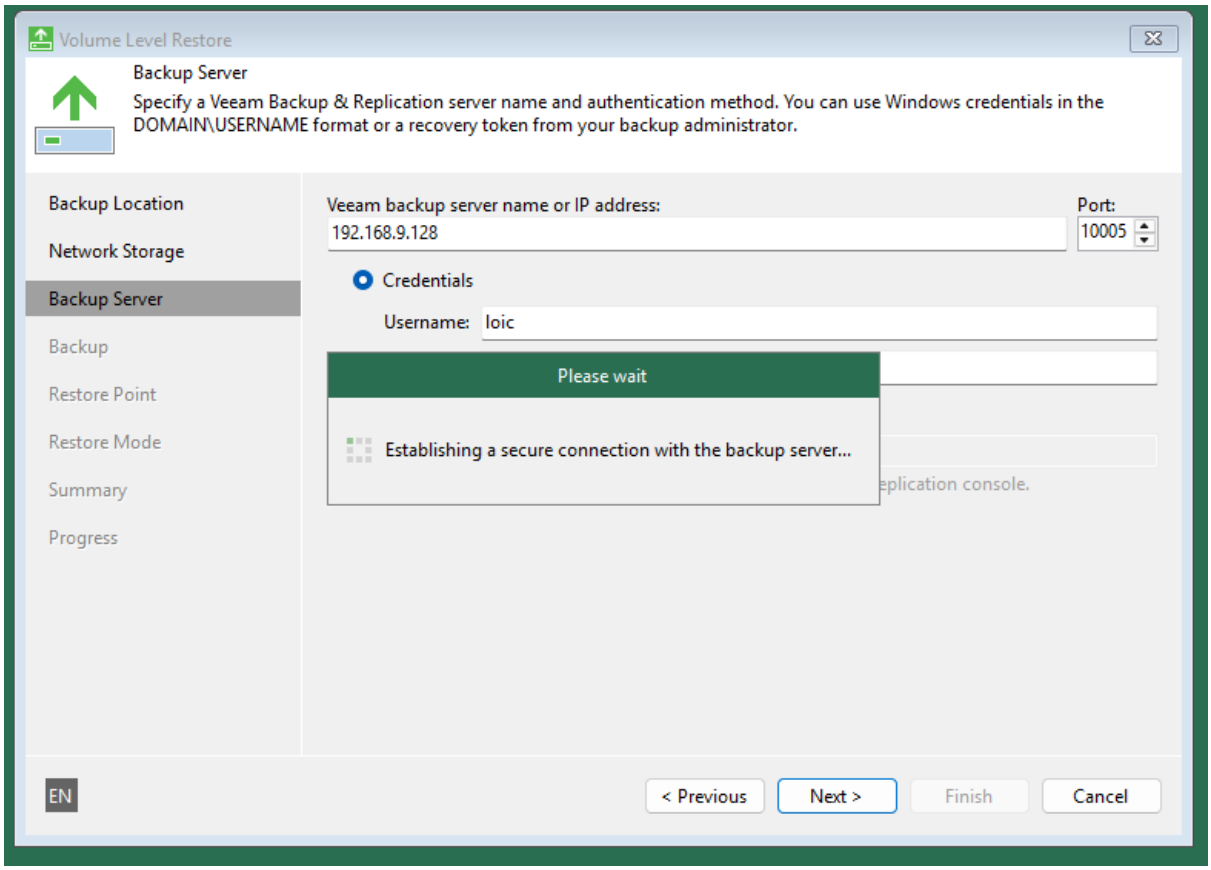


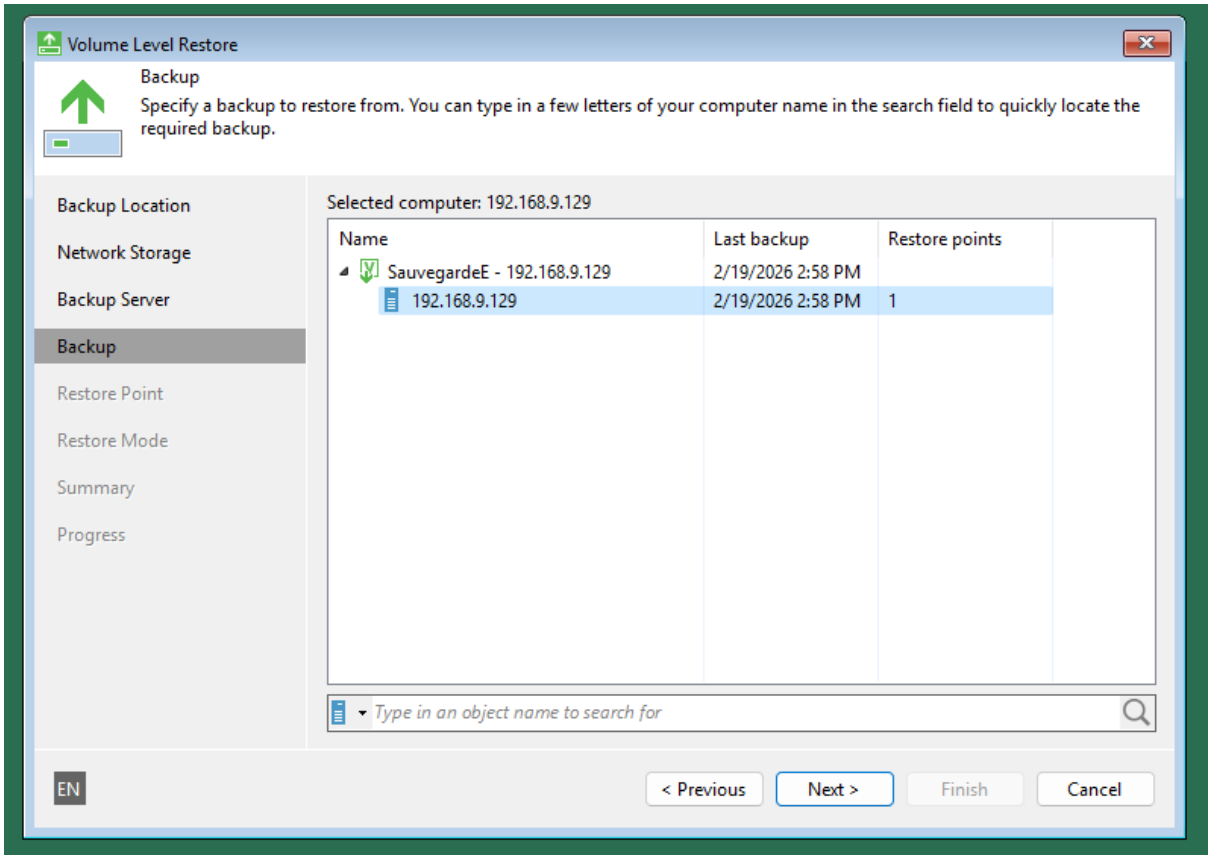
On sélectionne *Veeam backup repository*.

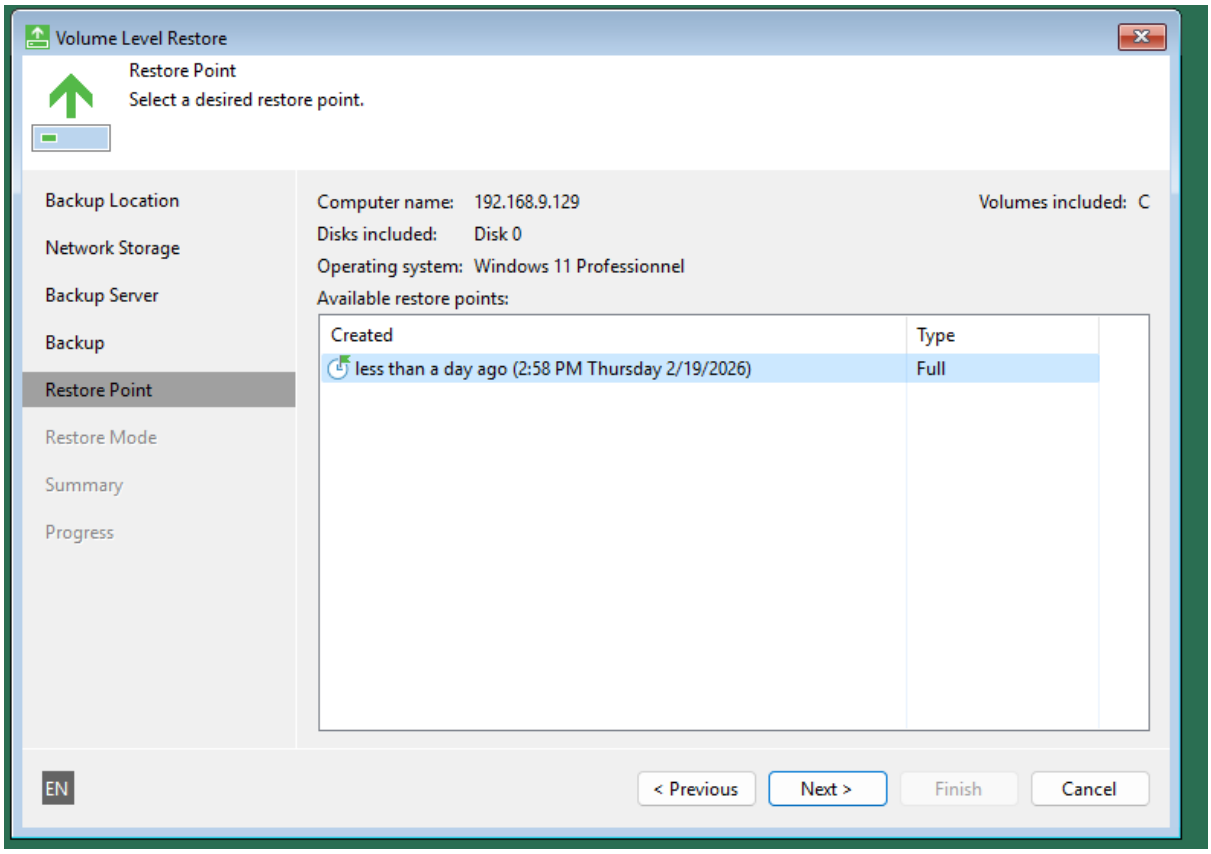


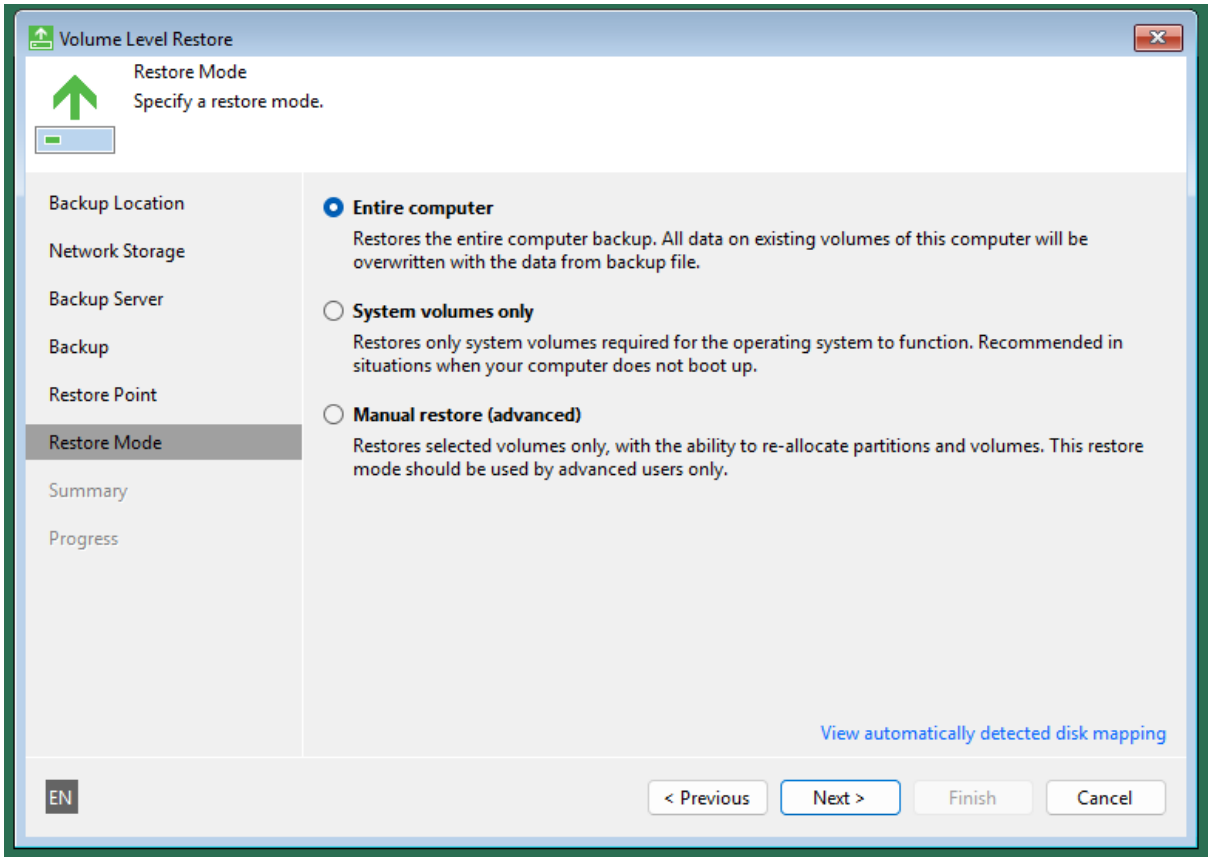
On indique l'adresse IP du Windows Server et les identifiants Windows.

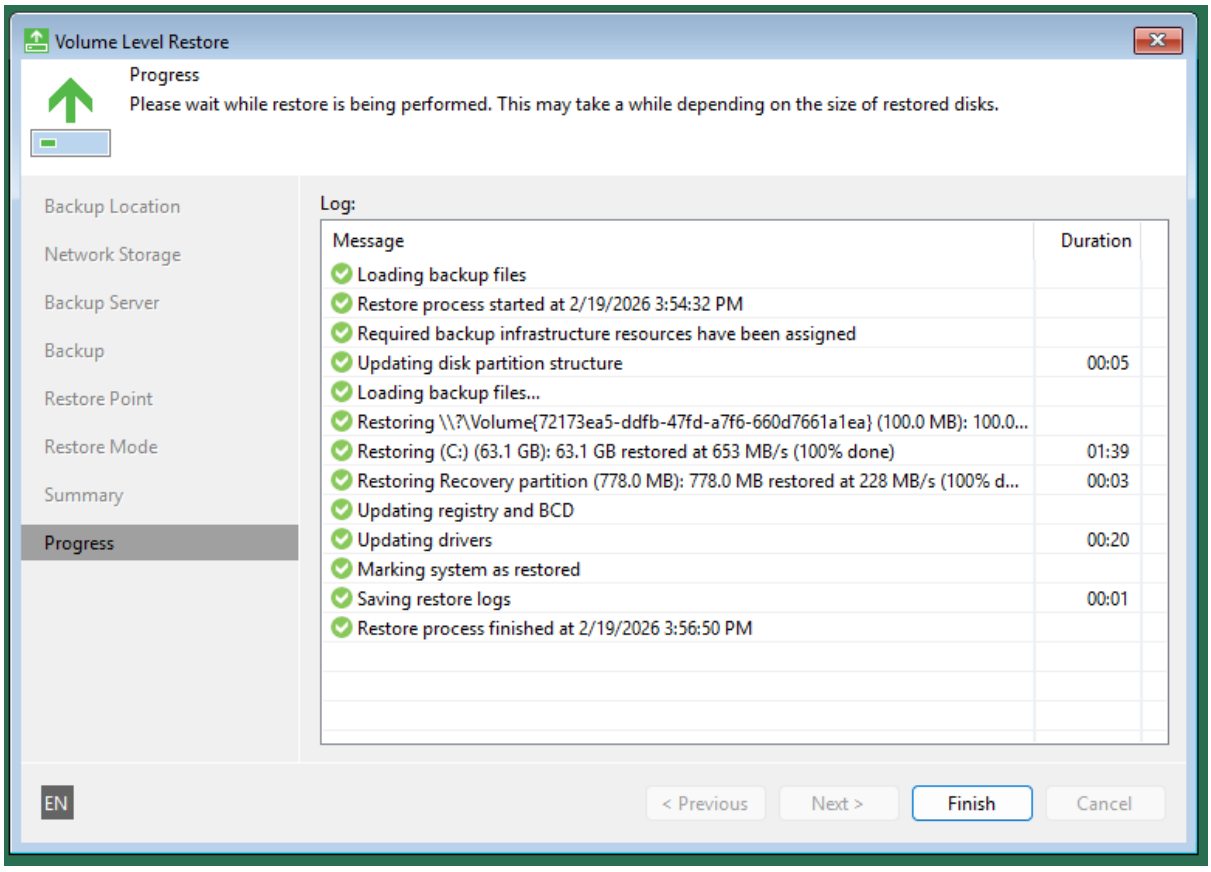





Volume Level Restore
Backup
Specify a backup to restore from. You can type in a few letters of your computer name in the search field to quickly locate the required backup.
Backup Location
Network Storage
Backup Server
Backup
Restore Point
Restore Mode
Summary
Progress
Selected computer: 192.168.9.129
Name
Last backup
Restore points
SauvegardeE - 192.168.9.129
2/19/2026 2:58 PM
192.168.9.129
2/19/2026 2:58 PM
1
Type in an object name to search for
EN
< Previous
Next >
Finish
Cancel


Volume Level Restore
Restore Point
Select a desired restore point.
Backup Location
Network Storage
Backup Server
Backup
Restore Point
Restore Mode
Summary
Progress
Computer name: 192.168.9.129
Volumes included: C
Disks included: Disk 0
Operating system: Windows 11 Professionnel
Available restore points:
Created
Type
less than a day ago (2:58 PM Thursday 2/19/2026)
Full
EN
< Previous
Next >
Finish
Cancel





La machine redémarre, et Windows se lance.

Conclusion

Nous avons désormais un serveur Windows sur lequel Veeam est installé. Nous avons effectué une sauvegarde d'une machine, et nous avons créé un iso qui a permis de lancer Windows sur une machine vierge.